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Design of closed Brayton cycle power generation system for megawatt-scale space nuclear reactor

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ABSTRACT

With advancements in aerospace technology, nuclear fission reactors have become essential for space power systems. However, the harsh space environment, complex structures, and varied choices of materials and working fluids pose significant design challenges. To facilitate the realization of high-power space nuclear propulsion systems, this paper proposes a universal research framework consisting of six modules: control, material properties, structural design, thermodynamic models, individual component/coupled system analysis, and feedback adjustment. A closed Brayton cycle is employed, using a He-Xe mixed gas as the working fluid, and the inputs are categorized into static and dynamic parameters. Corresponding models are developed with Python, Refprop, and simulation software, coupled into a model of space nuclear reactor closed Brayton cycle power generation system (SNRCBC). The steady-state simulations for individual components are calibrated, an eight-stage startup scheme is proposed, and transient simulations of the startup process are conducted, verifying the accuracy and correctness of both individual component models and the coupled system model. Finally, simulations are performed under four operating conditions: variable reactivity, variable shaft speed, variable working fluid flow, and variable load. The simulation results show that under steady-state conditions, the reactor core output power reaches 3.248 MW, the generated power is 1.072 MW, and the thermoelectric conversion efficiency is 33.03%. In the transient simulation under varying operating conditions, the system model demonstrates good self-regulation capabilities. This work provides valuable insights for the conceptual design of space nuclear reactor power generation systems.

1. Introduction

With the advancement of space technology, tasks like satellite communication and deep space exploration require power systems with higher output and longer lifespans to suit unique operational environments [1,2]. Nuclear energy, with its high energy density, longevity, and adaptability to extreme conditions, has become the preferred choice [3,4]. As illustrated in Fig. 1, the International Atomic Energy Agency (IAEA) categorizes space power sources into four types: chemical, solar, radioisotope, and space nuclear reactors power systems (SNRPS) [5,6]. Chemical power sources, while offering a wide power range, are limited by fuel capacity, resulting in short lifespans and poor sustainability. Solar power systems, widely used in space missions, provide longer lifespans but face instability due to factors like spacecraft position and light intensity, especially in deep-space missions where power rapidly declines with distance. Radioisotope systems, powered by radioactive decay, are stable and long-lasting but offer low power output. In contrast, SNRPS generate heat via nuclear fission, boasting high temperatures, independence from oxygen, immunity to light variations, and prolonged operation, making them highly promising for future missions [7,8].

Space nuclear power research began in the mid-20th century, led by the U.S. and the Soviet Union. In 1955, the U.S. launched the 'Rover' Project, later evolving into the NERVA (Nuclear Engine for Rocket Vehicle Application) program, which focused on nuclear thermal propulsion [9]. This method used heat from nuclear reactors to pressurize and heat propellants, expelling them through nozzles to generate thrust. By 1965, the U.S. launched the first spacecraft equipped with a nuclear thermoelectric power system (SNAP-10 A) for unmanned deep-space exploration [6,10]. In 1976, the U.S. successfully applied the multi-hundred watt radioisotope thermoelectric generator (MHW-RTG) to the communication satellites. The following year, NASA launched the 'Voyager' probes, also equipped with MHW-RTGs, which remain operational in space to this day [11]. During the same period,

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Nomenclature

P_f	Fission power of the reactor (MW)	ρ	Total reactivity ($)
ρ_0	Initial reactivity ($)	D	Hydraulic diameter (m)
C_i	Concentration of delayed neutron precursors	μ	Dynamic viscosity (Pa s)
T_f	Core temperature (K)	u	Velocity (m/s)
T_{f0}	Initial core temperature (K)	T_{out}	Core outlet temperature (K)
T_{in}	Core inlet temperature (K)	P_{TAC}	Surplus power (MW)
n	Neutron density (cm^2 s)	Σ_f	Macroscopic fission cross-section (cm^{-1})
E_f	Fission energy release (MeV)	V_f	Core fuel volume (m^3)
K_f	Proportionality coefficient	Pr	Prandtl number
$\dot{m}$	Mass flow rate of lithium (kg/s)	$\dot{m}_h$	Mass flow rate on hot side (kg/s)
f_{rc}	Compressor pressure ratio performance curve	$f_{\eta c}$	Compressor efficiency performance curve
f_{rt}	Turbine expansion ratio performance curve	$f_{\eta t}$	Turbine efficiency performance curve
$\dot{m}_c$	Compressor mass flow rate (kg/s)	$\dot{m}_t$	Turbine mass flow rate (kg/s)
N_{rpm}	Shaft speed (rpm)	$T_{c,in}$	Compressor inlet temperature (K)
$T_{c,out}$	Compressor outlet temperature (K)	$P_{c,in}$	Compressor inlet pressure (Pa)
$P_{c,out}$	Compressor outlet pressure (Pa)	$T_{t,in}$	Turbine inlet temperature (K)
$T_{t,out}$	Turbine outlet temperature (K)	$P_{t,in}$	Turbine inlet pressure (Pa)
$P_{t,out}$	Turbine outlet pressure (Pa)	$T'_{t,out}$	Ideal turbine outlet temperature (K)
φ_{1-2}	Average adiabatic coefficient at turbine inlet and outlet	$(1-2)$	Actual expansion work process
$(1-2')$	Isentropic expansion work process	$(4-5)$	Actual compression work process
$(4-5')$	Isentropic compression work process	$\overline{C}_{p,4-5}$	Average specific heat capacity during compression (J/kg K)
P_T	Expansion power of turbine (MW)	P_C	Compression power of compressor (MW)
Q_h	Heat released by working fluid on hot side (J)	Q_c	Heat absorbed by working fluid on cold side (J)
$\overline{C}_{p,c}$	Average specific heat capacity on cold side (J/kg K)	$T_{h,o}$	Hot fluid outlet temperature (K)
$\overline{C}_{p,h}$	Average specific heat capacity on hot side (J/kg K)	$T_{h,i}$	Hot fluid inlet temperature (K)
$\overline{B}$	Second-order Virial coefficient (He-Xe gas)	$\overline{C}$	Third-order Virial coefficient (He-Xe gas)
Z	Compressibility factor	x_1	Mole fraction of He
x_2	Mole fraction of Xe	M_1	Molar mass of He
M_2	Molar mass of Xe		
$T_{c,i}$	Cold fluid inlet temperature (K)	SNRPS	Space nuclear reactors power systems
$T_{c,o}$	Cold fluid outlet temperature (K)	MBL	Multiple Brayton loops
C	Heat capacity rate (W/K)	TAC	Turbine-Alternator-Compressor Model
C_h	Heat capacity rate on hot side (W/K)	TFE	The thermionic fuel element
C_c	Heat capacity rate on cold side (W/K)	SNPS-DBL	Space nuclear power system double Brayton loops
Nu	Nusselt number	He-Xe	Helium-xenon mixture gas
M_{rad}	Mass of the radiator (kg)	SNRCBC	Space nuclear reactor closed Brayton cycle power generation system
$C_{p,rad}$	Specific heat capacity of radiator (J/kg K)	CBC	Closed Brayton cycle
T_{rad}	Temperature of radiator (K)		
A_{rad}	Heat dissipation area of radiator (m^3)		
ϵ	Emissivity		
σ	Stefan–Boltzmann constant		

the Soviet Union developed space nuclear power systems, deploying BUK and TOPAZ-I on RORSAT satellites and testing the TOPAZ-II prototype [12]. However, the Soviet Union's technical approach differed from that of the U.S. The Soviets mainly utilized static power generation in nuclear reactors, which involves no mechanical moving parts during electricity generation. This method, based on thermoelectric principles, is less complex but offers lower efficiency, making it suitable for small power applications. Conversely, dynamic power generation incorporates moving mechanical components, offering higher efficiency and greater output but with increased design complexity [13]. In 1983, the U.S. initiated the SP-100 and SNTP programs for high-power systems, but these ended in 1993 [14]. Subsequent efforts focused on RTG advancements, like the GPHS-RTG, applied in missions such as 'Galileo', 'Ulysses', and 'Cassini' [15,16]. Meanwhile, the Soviet Union's research stalled after its collapse. In the early 21st century, the U.S. briefly pursued the 'Prometheus' program, but influencing later advancements [17]. China and the EU focused on RTGs, while Russia resumed high-power nuclear reactor research around 2010 [18,19]. Currently, space nuclear power systems are still in early stages globally. Russia plans reactor-powered spacecraft tests by 2030, the EU targets megawatt-class systems by 2040, and China aims for nuclear propulsion by 2045 [20]. Most space nuclear power programs in the past have failed, with the few successful applications being primarily low-power nuclear systems based on RTGs. The timeline of space nuclear power development is shown in Fig. 2.

To realize high-power space nuclear propulsion, integrating dynamic power conversion with fission reactors is essential. The dynamic power generation technologies in nuclear reactor power systems primarily include the Rankine cycle, Stirling cycle, and Brayton cycle [21,

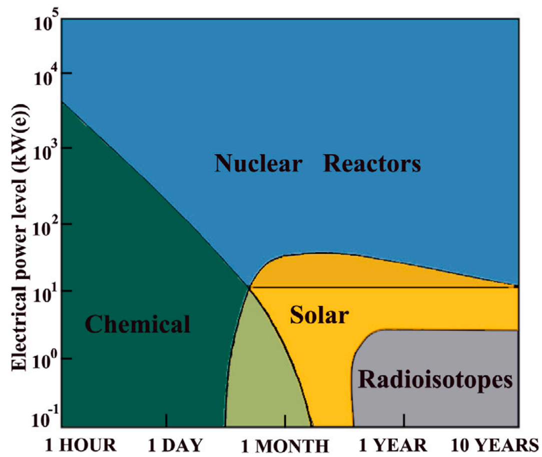


Fig. 1. Types of Space Power Sources and Applications [5].

22]. The Brayton cycle excels at handling high power demands, is well-suited for high-temperature and high-pressure conditions, offers fast dynamic response, and features a compact structure, making it ideal for space nuclear reactor applications [23]. As shown in Fig. 3, the space nuclear reactor closed Brayton cycle power generation system (SNRCBC) is primarily composed of three heat exchange loops. The first loop consists of the reactor core and the heater. The fission heat generated by the reactor is carried by the coolant flowing through the core to the heater, where it exchanges heat with the working fluid of the second loop before returning to the reactor core, continuing in a cyclic process. The second loop comprises the turbine, compressor, recuperator, and heater. The working fluid of the second loop, after being heated by the coolant from the first loop in the heater, flows through the turbine to expand, converting the thermal energy of the working fluid into mechanical energy. The turbine, compressor, and alternating current generator are typically mounted on a single shaft for coaxial rotation, converting mechanical energy into electrical power. The heated working fluid, after passing through the turbine outlet, enters the recuperator, where it heats the cold working fluid entering the recuperator from the compressor, before being cooled in the cooler and transformed into cold working fluid, which is then compressed in the compressor to complete the cycle. The third loop consists of the radiator and cooler. The coolant in the radiator exchanges heat with the hot working fluid flowing out of the recuperator in the cooler. The waste heat is carried back to the radiator and then expelled into outer space through fins [24,25]. The system's T-S diagram (Temperature-Entropy) is shown in Fig. 4.

In recent years, comprehensive modeling of the SNRCBC — from transient start-up through steady-state operation to varying working conditions under real-world scenarios — has remained relatively scarce [26–28]. El-Genk et al. [29] developed the TITAM lumped-parameter model for the on-orbit startup of the static thermionic reactor TOPAZ. Astrin et al. [30] used Simulink to model the TOPAZ-II startup process. Steven A.W. et al. [31–34] proposed the RPCSIM dynamic model to study the startup and transient safety of a liquid-metal-cooled Brayton reactor. Yuan et al. [35] introduced the TAPIRS code for thermosyphon-cooled systems, while El-Genk et al. [36,37] developed Simulink models for low-power thermoelectric systems, focusing on startup and load-following dynamics. In addition, Wu et al. [38] used a self-developed thermal-hydraulic analysis program to assess the transient safety of the gas-cooled reactor and the performance of the coupled Brayton cycle system. Wu et al. [39] developed a dynamic model for a 100 kW lithium-cooled reactor Brayton power system using the RESYS code. Ge et al. [40] used the developed transient analysis program TACSNR to calculate and analyze the transient process of the design startup scheme, proposing a multiphase startup plan based on reactor startup followed by Brayton cycle system closure. This approach minimizes the auxiliary motor power required for turbine startup. Ma et al. [41,42] studied the space nuclear power system (SNPS) with multiple Brayton loops (MBL), establishing a dynamic model for SNPS-MBL. They investigated the characteristics of the dual Brayton loop (SNPS-DBL) after the loss of load in both single and dual Brayton loops (BL). Zhang et al. [43] developed the SNRPS analysis software, which integrates a megawatt-class gas-cooled reactor. The primary objective was to simulate and analyze mechanical failures associated with MFOBL1 and RIA accidents, while simultaneously validating the system's self-regulating capabilities under accident conditions. Zhang et al. [44] proposed an ACMIR integrated space nuclear reactor scheme centered on a Stirling energy converter, employing 3D CFD and dynamic mesh methods to simulate thermal-fluid behavior with a focus on local structural optimization. Miao et al. [45] developed a recompression Brayton cycle model using N_2O-He mixtures for deep space exploration, highlighting improvements in thermal efficiency and system mass via pressure and split ratio adjustments, though lacking dynamic control and full-condition modeling. Lu et al. [46] studied an S-CO_2 Brayton-CHP system for Mars base power and heat supply, optimizing efficiency and mass through a CPO-SVR-COA hybrid algorithm, yet focused mainly on static performance without deployable control logic or startup simulation. Xu et al. [47,48] investigated the impact of working fluid characteristics under non-ideal gas conditions on the power generation performance of the He-Xe Brayton cycle in space lithium-cooled reactors. However, a comprehensive study of the system's response characteristics was not carried out. In addition, Liu et al. [49–51] conducted studies on the efficiency, safety, and structural configuration of SNRCBC from different perspectives, such as individually optimizing certain components, establishing a comprehensive structural model for a specific optimization objective, and developing evaluation models to assess the impact of working fluids.

A systematic comparison between the SNRCBC model developed in this study and representative works from existing literature is presented in Table 1. While prior studies have made valuable progress in the modeling and transient analysis of space nuclear reactors, several important limitations remain. Most notably, existing research tends to emphasize structural or thermal-fluid simulations at the component level, often relying on CFD tools or static system models [26–28]. Although dynamic models such as TITAM [29], RPCSIM [31–34], and TAPIRS [35] have been developed, they are typically tailored to specific configurations or focused on narrow operational scenarios, such as single transient events or simplified startup sequences. In addition, many studies do not fully integrate the coupled behavior of the reactor and Brayton power system, or lack generalizability due to rigid parameter settings and closed-source platforms. Efforts such as those by Wu et al. [38,39] and Ge et al. [40] offer more refined transient analysis, but remain limited in scope and flexibility. Overall, there is a lack of a comprehensive, modular simulation framework that captures the multiphysics and multi-stage behavior of SNRCBCs under realistic operating conditions and supports broad mission adaptability. To address these gaps, this work proposes a transferable, Python-based simulation platform designed for general-purpose modeling of space nuclear Brayton systems, as shown in Fig. 5. The framework couples steady-state and transient behavior across modular components, incorporates customizable material and fluid property libraries, and supports realistic control logic through an eight-stage, rule-based startup sequence. This approach enables system-level exploration and optimization of SNRCBC designs, making it a more universally applicable megawatt-class power system for a wide range of future space mission scenarios, while achieving a thermal efficiency of up to 33.03%. Distinct from existing studies, the primary contributions of this paper are as follows:

- Unlike prior studies that focus on isolated system components or specific scenarios, this paper proposes a universal modular research framework consisting of six key modules: control,

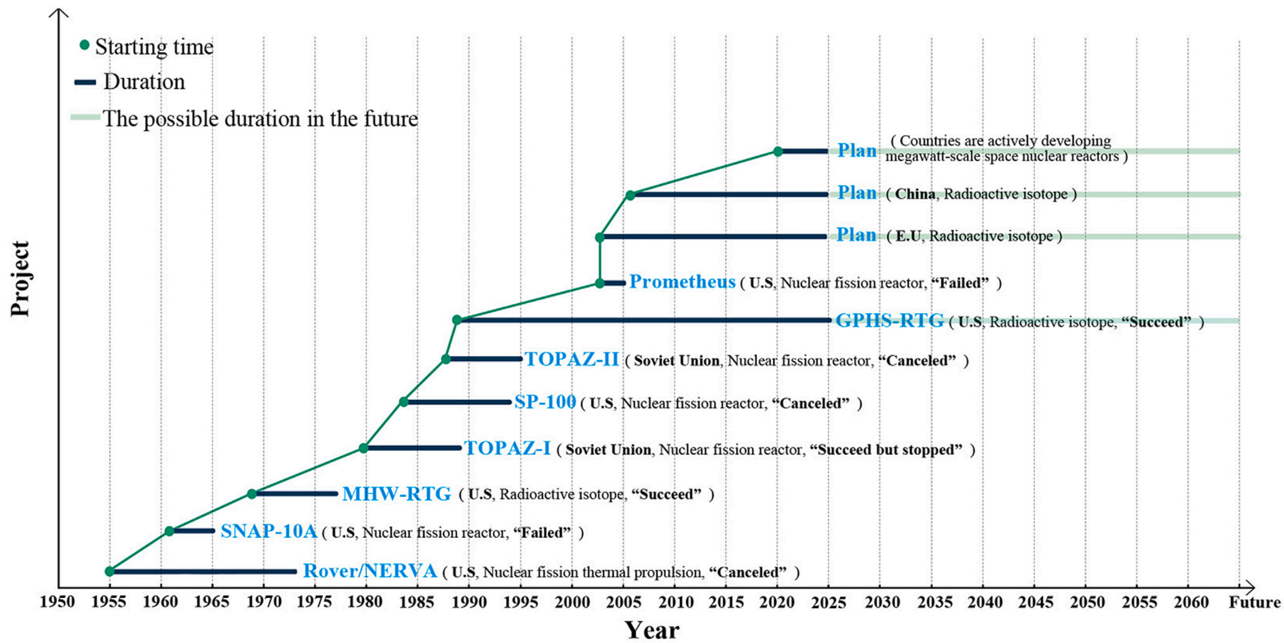


Fig. 2. Timeline of space nuclear power development.

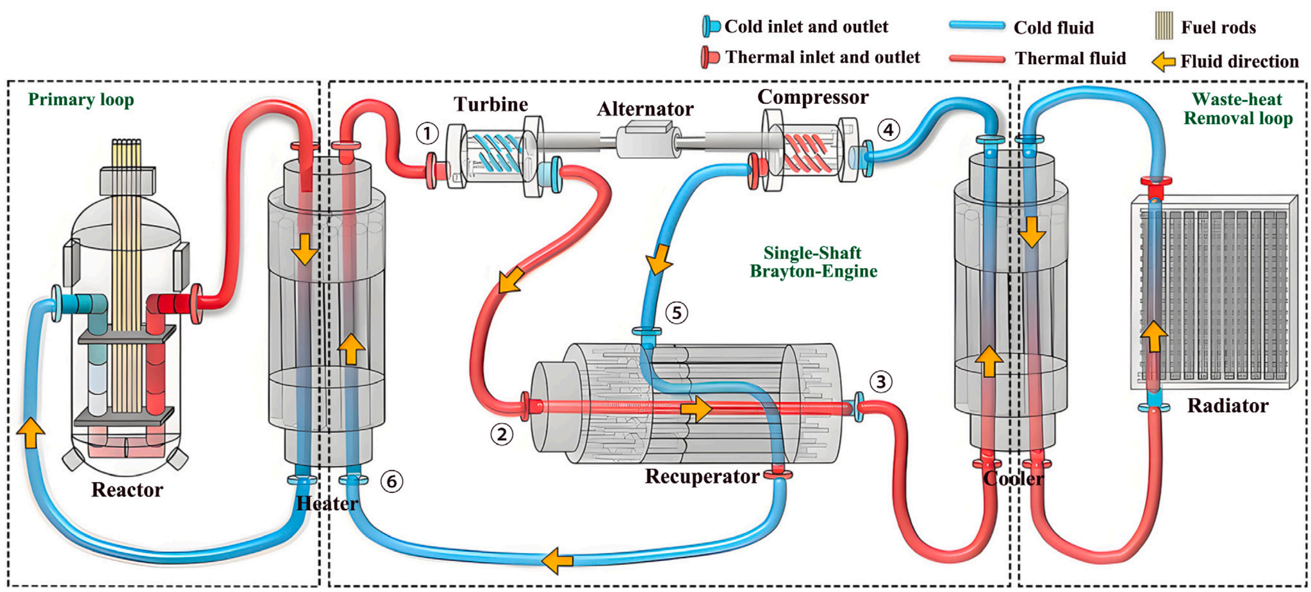


Fig. 3. A schematic of the SNRCBC.

material properties, structural design, thermodynamic modeling, component/system coupling, and feedback adjustment. This structure enables flexibility across both static and dynamic input parameters—a novelty not addressed in existing studies such as [38–51].

- Most existing studies focus either on steady-state performance or simplified transient events. This paper uniquely combines component-level steady-state calibration with system-level transient simulations, including startup and four dynamic conditions (reactivity, shaft speed, fluid flow, and load). As shown in Table 1, previous studies have not simultaneously covered such a comprehensive dynamic range, making it difficult to demonstrate the completeness of their models.
- A novel eight-phase startup process governed by a rule-based control module is proposed, drawing from small-scale Brayton startup literature [40,48] but extended to realistic engineering conditions, including solid-state lithium coolant thawing, thermal lag considerations, and safety-driven phase transitions.
- The model incorporates a dedicated material property package including non-ideal gas models for He–Xe using Virial coefficients, as well as metallic coolant (Li, NaK) and structure materials (Ni alloy, Al alloy) [52]. Such an extensible package integrating thermophysical property details specific to space reactors was not available in previous SNRCBC simulations.

The remainder of this paper is organized as follows: In Section 2, we introduce the modules in the SNRCBC design framework, and corresponding models were established, followed by the presentation of simulation results and analysis in Section 3. Finally, we conclude the paper in Section 4.

Table 1
Comparison of related work.

Related work	Analysis mode	The model is transferable	Power generation capacity	Primary cycle coolant	Thermal circuit	Changes in operating conditions			
						Reactivity	Shaft speed	Load	Coolant flow
Ref [38]	Transient	NO	Megawatt	He-Xe	Brayton	✓	✓		✓
Ref [39]	Transient	NO	Kilopower	He-Xe	Brayton	✓	✓		✓
Ref [41]	Equilibrium	NO	Kilopower	He-Xe	Brayton			✓	
Ref [43]	Equilibrium	NO	Kilopower	He-Xe	Brayton	✓			✓
Ref [42]	Transient	NO	Kilopower	He-Xe	Brayton		✓		
Ref [44]	Transient	NO	Kilopower	He	Stirling				✓
Ref [45]	Equilibrium	NO	Kilopower	N_2O-He	Brayton				
Ref [46]	Transient	NO	Kilopower	S-CO_2	Brayton			✓	
This paper	**Transient & Equilibrium**	**YES**	Megawatt	He-Xe	Brayton	✓	✓	✓	✓

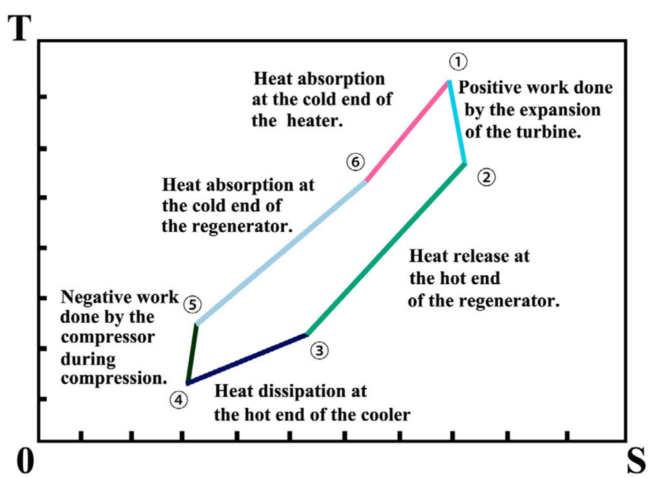


Fig. 4. T-S diagram of system.

2. System modules

Based on the partial module modeling of the research report [53], this system is mainly divided into the following several parts: the Reactor Core Model, which serves to generate heat; the Turbine-Alternator-Compressor Model (TAC), which facilitates power generation; the Heat Exchange Model, responsible for thermal exchange; and the Material Property Package, which characterizes the physical properties of the materials and working fluids. Each of these models can be used independently to study local characteristics, or they can be coupled to examine the overall properties of the closed Brayton cycle system. Additionally, both transient and steady-state models have been established.

2.1. Reactor core model

The thermionic fuel element (TFE) where the fission and thermionic reactions occur is the basic unit of a space thermionic nuclear reactor, arranged in the moderator, as shown in Fig. 6. The analysis of reactor power, temperature and reactivity mainly uses the reactor point reactor dynamics equation [53]. This paper utilizes six groups of delayed neutron point reactor kinetics equations to describe the core model, as shown in Formulas (1) to (6).

$$\frac{d}{dt}P_f = \frac{1}{\Lambda}\cdot(\rho - \sum_{i=1}^{6}\beta_i)\cdot P_f + \sum_{i=1}^{6}\lambda_i C_i \tag{1}$$

$$\frac{dC_i}{dt} = \frac{\beta_i}{\Lambda}P_f - \lambda_i C_i, i = 1,2,3,\ldots,6 \tag{2}$$

$$\beta_i = \beta \cdot \alpha_i \tag{3}$$

Table 2
Parameters of the reactor core.

Parameter	Value	Parameter values		
		i	λ_i (s^{-1})	α_i
β	0.0064	1	0.0127	0.0380
Λ (s)	10^{-4}	2	0.0317	0.0213
$\alpha_f(dk/k/K)$	$-1.28 \cdot 10^{-5}$	3	0.1150	0.1880
$\alpha_c(dk/k/K)$	$-1.67 \cdot 10^{-5}$	4	0.3110	0.4070
k_f (K/J)	$3.425 \cdot 10^{-5}$	5	1.4000	0.1280
Γ_f (s^{-1})	0.0534	6	3.8700	0.0260

$$\rho = \rho_0 + \alpha_f \cdot (T_f - T_{f0}) + \alpha_c \cdot (T_{out} - T_{in}) \tag{4}$$

where P_f represents the fission power of the reactor. Λ denotes the prompt neutron generation time. ρ and ρ_0 respectively represent the total reactivity and initial reactivity. λ_i, β_i, and C_i respectively represent the delayed neutron decay constants, the fraction of delayed neutrons, and the concentration of delayed neutron precursors. β is sum of β_i, α_i denotes the relative delayed neutron yield fraction. α_f is the fuel temperature negative feedback coefficient, α_c is the coolant temperature negative feedback coefficient, T_f represents the core temperature, T_{f0} denotes the initial core temperature, T_{out} refers to the core outlet temperature, and T_{in} stands for the core inlet temperature.

$$P_f = K_f \cdot n \tag{5}$$

$$K_f = \Sigma_f \cdot E_f \cdot V_f \tag{6}$$

where n denote the neutron density, while Σ_f, E_f, and V_f represent the macroscopic fission cross-section, the fission energy release, and the core fuel volume, respectively. Together, they form the proportionality coefficient K_f. Parameter values as outlined in Table 2 [54].

The lumped parameter method is used to describe the relationship between heat transfer and temperature in the reactor [51]. To simplify, the convective heat transfer is calculated using the average temperature T_{ave} of the reactor fuel and liquid lithium coolant. The heat transfer and temperature variations in the core are determined by Formulas (7) to (9).

$$\frac{d}{dt}T_f = k_f \cdot P_f - \Gamma_f(T_f - Tave) \tag{7}$$

$$P_{th} = \dot{m} \cdot C_{p,L}(T_{out} - T_{in}) \tag{8}$$

$$T_{ave} = \frac{1}{2} \cdot (T_{out} + T_{in}) \tag{9}$$

k_f denotes the heat capacity of the fuel element, Γ_f represents the time constant for heat transfer from the fuel to the coolant, P_{th} is the thermal power transferred from the reactor core to the lithium coolant, $C_{p,L}$ is specific heat capacities at constant pressure of the lithium, and $\dot{m}$ refers to the mass flow rate of the lithium.

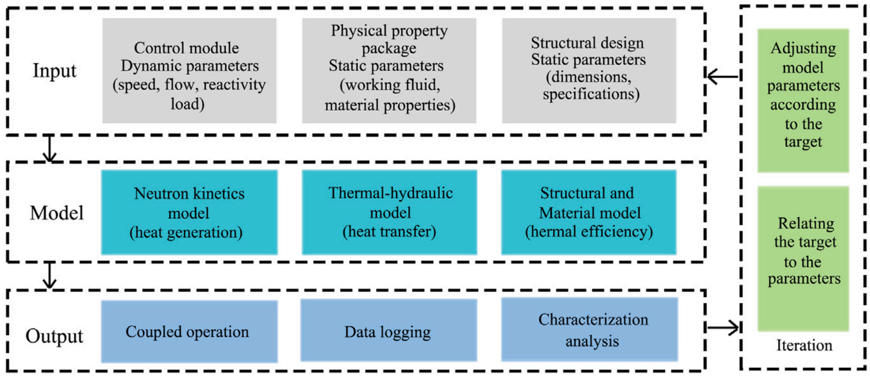


Fig. 5. SNRCBC design framework.

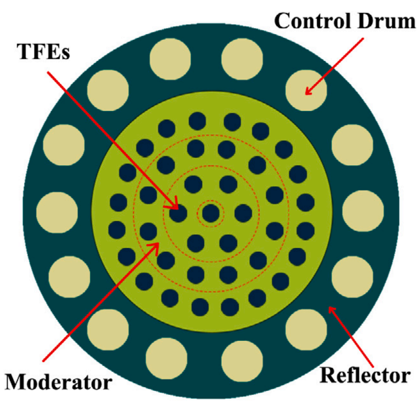


Fig. 6. Cross-sectional diagram of the reactor core.

2.2. TAC model

The turbine, alternator, and compressor are coaxially arranged and operate at the same speed. The compressor and turbine are shown in Fig. 7 [55], with the model descriptions provided in Formulas (10) to (14).

$$r_t = \frac{P_{t,in}}{P_{t,out}} = f_{rt}(T_{t,in}, P_{t,in}, \dot{m}_t, N_{rpm}) \tag{10}$$

$$r_c = \frac{P_{c,out}}{P_{c,in}} = f_{rc}(T_{c,in}, P_{c,in}, \dot{m}_c, N_{rpm}) \tag{11}$$

$$\dot{m}_t = \dot{m}_c = f_{mt}(T_{t,in}, P_{t,in}, r_t, N_{rpm}) \tag{12}$$

$$\eta_t = f_{\eta t}(T_{t,in}, P_{t,in}, \dot{m}_t, N_{rpm}) \tag{13}$$

$$\eta_c = f_{\eta c}(T_{c,in}, P_{c,in}, \dot{m}_c, N_{rpm}) \tag{14}$$

The functions f_{rc} and $f_{\eta c}$ represent the performance curve relationships for the compressor's pressure ratio r_c and efficiency η_c, while f_{rt} and $f_{\eta t}$ represent the performance curve relationships for the turbine's expansion ratio r_t and efficiency η_t. These functions are dependent on mass flow rate $\dot{m}_c$, $\dot{m}_t$ and shaft speed N_{rpm}. Then $T_{c,in}, T_{c,out}, P_{c,in}, P_{c,out}$ are the compressor's inlet and outlet temperatures and pressures, while $T_{t,in}, T_{t,out}, P_{t,in}, P_{t,out}$ are the turbine's inlet and outlet temperatures and pressures. In this case, The working fluid expands and does work in the turbine, with the ideal outlet temperature $T'_{t,out}$ given by:

$$T'_{t,out} = T_{t,in} \cdot \left(\frac{P'_{t,out}}{P_{t,in}}\right)^{\frac{\gamma_t - 1}{\gamma_t}} = r_t^{\varphi_{1-2}} \tag{15}$$

where $\varphi_{1-2} = \frac{\gamma_t - 1}{\gamma_t}$ is the average adiabatic coefficient at the turbine inlet and outlet, and its value depends on the specific heat capacity of the working fluid, while $(1-2)$ represents the actual expansion work process. $P'_{t,out}$ is the ideal outlet pressure. $\gamma_t = \hat{C}_p/\hat{C}_v$ is the specific heat ratio of working fluid, see Section 2.4, formulas (33) to (36). Then, formula (13) can be expressed as:

$$\eta_t = \frac{\overline{C}_{p,1-2}(T_{t,in} - T_{t,out})}{\overline{C}_{p,1-2'}(T_{t,in} - T'_{t,out})} \tag{16}$$

where $(1-2')$ denotes the isentropic expansion work process. $\overline{C}_{p,1-2}$ signifies the average specific heat capacity at constant pressure for the working fluid as it moves from the turbine inlet to the outlet. The work done during the expansion in the turbine can be calculated:

$$W_T = \dot{m}\overline{C}_{p,1-2}(T_{t,in} - T_{t,out}) \tag{17}$$

Similarly, the work consumed during the compression of the working fluid in the compressor, the ideal outlet temperature of the compressor $T'_{c,out}$, is given by:

$$T'_{c,out} = T_{c,in} \cdot r_c^{\varphi_{4-5}} \tag{18}$$

where $\varphi_{4-5} = \frac{\gamma_c - 1}{\gamma_c}$ is the average adiabatic coefficient at the compressor inlet and outlet. γ_c is also the specific heat ratio of working fluid. Then, formula (14) can be expressed as:

$$\eta_c = \frac{\overline{C}_{p,4-5'}(T'_{c,out} - T_{c,in})}{\overline{C}_{p,4-5}(T_{c,out} - T_{c,in})} \tag{19}$$

where $(4-5)$ represents the actual compression work process, while $(4-5')$ denotes the isentropic compression work process. $\overline{C}_{p,4-5}$ represents the average specific heat capacity at constant pressure for the working fluid during the process from the compressor inlet to the outlet. The work consumed during the compression process in the compressor can be calculated:

$$W_C = \dot{m}\overline{C}_{p,4-5}(T_{c,out} - T_{c,in}) \tag{20}$$

In the thermodynamic model, their performance is described by isentropic efficiency. However, for dynamic system simulations, it is essential to account for impeller speed and real-time inlet conditions, making the characteristic flow curves of the compressor and turbine more appropriate for performance evaluation [48,56]. The characteristic curves used are listed in Figs. 8 and 9.

For the dynamic analysis, since the turbine, alternator, and compressor are arranged coaxially in the TAC model, it is necessary to consider the relationship between the shaft speed N_{rpm}, rotor torque I_{TAC}, and component load P_{load}, as shown in Formulas (21) and (22).

$$P_{TAC} = P_T - P_C - P_{load} \tag{21}$$

$$\frac{d}{dt}N_{rpm} = \frac{900P_{TAC}}{\pi^2 I_{TAC} N_{rpm}} \tag{22}$$

where P_T and P_C represent the expansion power of the turbine and the compression power of the compressor, respectively, while P_{TAC} denotes the surplus power.

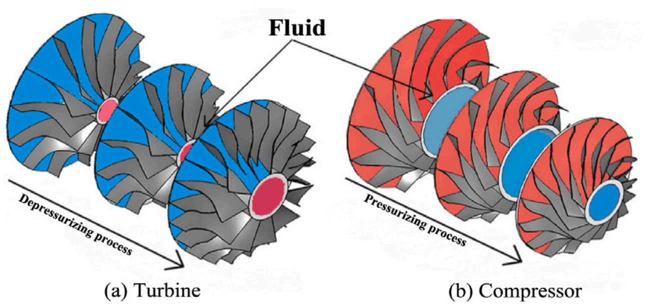


Fig. 7. Compressor and Turbine.

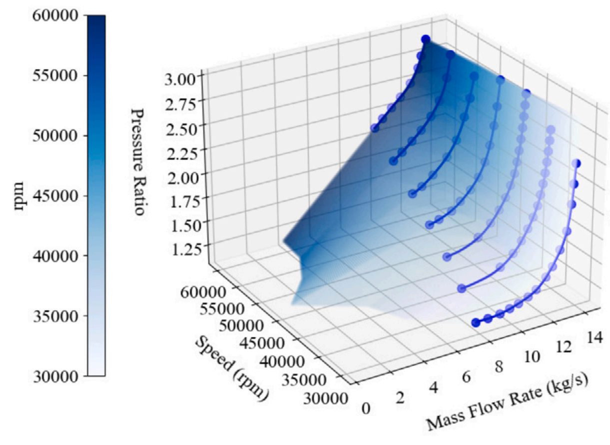


Fig. 8. The characteristic curves of turbine.

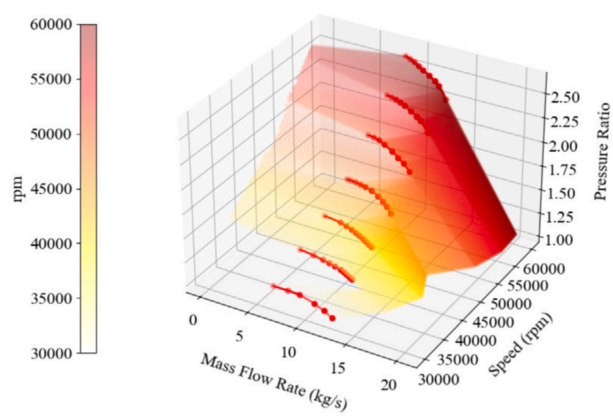


Fig. 9. The characteristic curves of compressor.

2.3. Heat exchanger model

The heat exchanger model is primarily divided into four models: heater, recuperator, cooler, and radiator. To facilitate efficient heat exchange while minimizing volume, the models of the heater, recuperator, and cooler are similar, all utilizing counterflow arrangements of printed circuit heat exchangers (PCHEs) [3,57] with straight flow channels, divided into hot and cold sides, as shown in Fig. 10. By calculating the heat transfer and heat exchange process of the fluid within the channels, combined with the energy balance equation, the model is established.

Firstly, in steady-state operation, the thermal load on both the cold and hot sides can be calculated using the mass flow rate and the inlet and outlet temperatures:

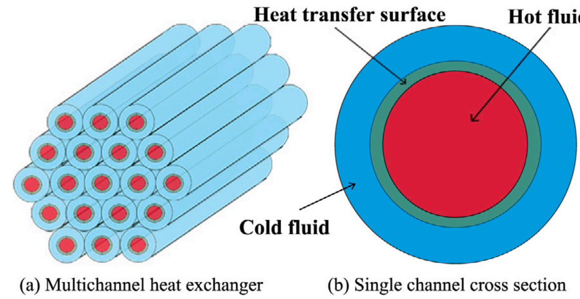


Fig. 10. Heat exchanger structure.

$$Q_h = \dot{m}_h \overline{C}_{p,h}(T_{h,i} - T_{h,o}) \tag{23}$$

$$Q_c = \dot{m}_c \overline{C}_{p,c}(T_{c,o} - T_{c,i}) \tag{24}$$

$$Q_h = Q_c \tag{25}$$

where Q_h and Q_c represent the heat released by the working fluid on the hot side and the heat absorbed by the working fluid on the cold side, respectively, $\overline{C}_{p,h}$ and $\overline{C}_{p,c}$ denote the average specific heat capacities of the working fluids on the hot and cold sides, respectively. $T_{h,i}$, $T_{h,o}$, $T_{c,i}$, and $T_{c,o}$ are the inlet and outlet temperatures of the hot and cold fluids. The heat capacity rate is denoted by C, with the heat capacity rate on the hot side given by $C_h = \dot{m}_h \overline{C}_{p,h}$ and the heat capacity rate on the cold side given by $C_c = \dot{m}_h \overline{C}_{p,c}$. The maximum heat transfer of the heater is then:

$$Q_{max} = C_{min}(T_{h,i} - T_{c,i}) \tag{26}$$

$$C_{min} = min(C_h, C_i) \tag{27}$$

The efficiency of the heater can thus be expressed as follows:

$$\epsilon_e = \frac{C_h(T_{h,i} - T_{h,o})}{C_{min}(T_{h,i} - T_{c,i})} = \frac{C_c(T_{c,o} - T_{c,i})}{C_{min}(T_{h,i} - T_{c,i})} \tag{28}$$

The transient model of the heater, can be described through the heat exchange between the cold-side fluid, the hot-side fluid, and the device itself.

For the hot side passage:

$$m_h C_{p,h} \frac{d\overline{T}_h}{dt} = \dot{m}_h C_{p,h}(T_{h,i} - T_{h,o}) - h_h A_1(\overline{T}_h - T_{HX}) \tag{29}$$

For the cold side passage:

$$m_c C_{p,c} \frac{d\overline{T}_c}{dt} = \dot{m}_c C_{p,c}(T_{c,i} - T_{c,o}) + h_c A_2(T_{HX} - \overline{T}_c) \tag{30}$$

For the heater solid layer:

$$M_{HX} C_{p,HX} \frac{dT_{HX}}{dt} = h_h A_1(\overline{T}_h - T_{HX}) - h_c A_2(T_{HX} - \overline{T}_c) \tag{31}$$

where m_h, m_c, and M_{HX} are the masses of the hot fluid, cold fluid, and heater, respectively. $C_{p,h}$, $C_{p,c}$, and $C_{p,HX}$ are the specific heat capacities at constant pressure for the hot fluid, cold fluid, and heater. $\overline{T}_h = \frac{1}{2}(T_{h,i} + T_{h,o})$, $\overline{T}_c = \frac{1}{2}(T_{c,i} + T_{c,o})$. Then T_{HX} is the heater temperature. The h_h and h_c are the heat transfer coefficients for the hot and cold fluids, and A_1 and A_2 are the heat transfer areas. Additionally, as shown in Fig. 10. (b), a circular tube heat exchange channel is used, with the inner pipe as the hot flow passage and the outer pipe as the

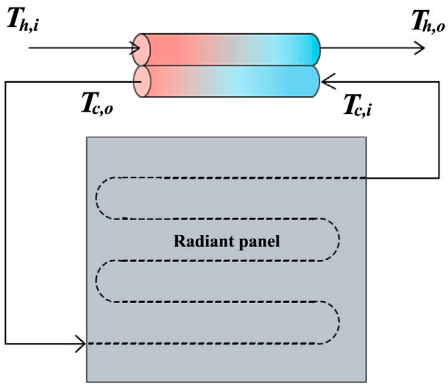


Fig. 11. Radiation cooling system.

cold flow passage. The fluid diameter inside is d_1, the heat transfer wall thickness is r, the inner diameter of the cold flow passage is $d_2 = d_1+2r$, and the outer diameter is d_3. The equivalent diameter of the hot flow passage is $D_h = d_1$, while the equivalent diameter of the outer cold flow passage D_c is given by:

$$D_c = \pi \cdot (d_3^2 - d_2^2)/(\pi \cdot d_2) \tag{32}$$

The model of the radiator differs from the three aforementioned heat exchangers in that the hot fluid in the heat channel carries heat into the radiator, where the heat is then dissipated to outer space through thermal radiation, rather than by convection heat transfer, as shown in Fig. 11. Therefore, the radiator model is represented as follows:

$$m_h C_{p,h} \frac{d\overline{T}_h}{dt} = \dot{m}_h C_{p,h}(T_{h,i} - T_{h,o}) - h_h A_1(\overline{T}_h - T_{rad}) \tag{33}$$

$$M_{rad} C_{p,rad} \frac{dT_{rad}}{dt} = h_h A_1(\overline{T}_h - T_{rad}) - A_{rad}\epsilon\sigma(T_{rad}^4 - T_{env}^4) \tag{34}$$

Here, M_{rad} is the mass of the radiator, $C_{p,rad}$ is the specific heat capacity at constant pressure of the radiator, T_{rad} is the temperature of the radiator, A_{rad} is the heat dissipation area of the radiator, ϵ is the emissivity of the object, and $\sigma = 5.67 \times 10^{-8}$ W/(m^2 K^4) is the Stefan–Boltzmann constant.

2.4. Material property package

The Material Property Model includes the thermophysical properties of the liquid lithium coolant in the first loop, He-Xe coolant in the second loop, Nak alloy coolant in the third loop, and materials such as nickel-based alloys, aluminum alloys, and stainless steel. To more accurately reflect the actual situation, the He-Xe coolant is simulated as a non-ideal gas. The physical properties of helium and xenon are mainly determined through the Refprop software using the lookup method and by calling Python. The calculation is carried out using the mixed estimation method from Ref. [47,58].

Firstly, the specific heat of a mixed gas primarily includes the molar specific heat at constant pressure $\hat{C}_p$ and the molar specific heat at constant volume $\hat{C}_v$. The molar specific heat at constant pressure, $C_{p,c}$ and $C_{p,h}$, for the He-Xe gas in the cold and hot side flow channels of the heat exchange model in formulas (23), (24), (29), (30) in Section 2.3 are given by:

$$\hat{C}_p = \hat{C}_p^0 + \hat{\rho} R_g[(\overline{B} - T\frac{d\overline{B}}{dT} - T^2\frac{d^2\overline{B}}{dT^2}) + \hat{\rho}(\overline{C} - \frac{T^2}{2}\frac{d^2\overline{C}}{dT^2})] + R_g T[(\overline{B} - T\frac{d\overline{B}}{dT}) + \hat{\rho}(2\overline{C} - T\frac{d\overline{C}}{dT})] \cdot \frac{\partial\hat{\rho}}{\partial T} \tag{35}$$

where $\hat{C}_p^0 = \frac{5}{2}R_g$, with $R_g = 8.314$ J/mol K being the gas constant, $\hat{\rho}$ is the molar density of the gas, T is the gas temperature, and $\overline{B}$ and $\overline{C}$ represent the second- and third-order Virial coefficients for the He-Xe gas, respectively. The calculation of $\frac{\partial\hat{\rho}}{\partial T}$ is as follows:

$$\frac{\partial\hat{\rho}}{\partial T} = -(\frac{\hat{\rho} + \overline{B}\hat{\rho}^2 + \overline{C}\hat{\rho}^3}{T} + \frac{d\overline{B}}{dT}\hat{\rho}^2 + \frac{d\overline{C}}{dT}\rho^3) \cdot (1 + 2\overline{B}\hat{\rho} + 3\overline{C}\hat{\rho}^3)^{-1} \tag{36}$$

The molar specific heat at constant volume and the specific heat ratio of He-Xe gas are calculated as follows:

$$\hat{C}_v = \hat{C}_v^0 - \hat{\rho} R_g T[(2\frac{d\overline{B}}{dT} + T\frac{d^2\overline{B}}{dT^2}) + \hat{\rho}(\frac{d\overline{C}}{dT} + \frac{T}{2}\frac{d^2\overline{C}}{dT^2})] \tag{37}$$

$$\gamma = \frac{\hat{C}_p}{\hat{C}_v} \tag{38}$$

where $\hat{C}_v^0 = \frac{3}{2}R_g$. $\hat{\rho} = \rho_{mix}/M_{mix}$, where ρ_{mix} is the mass density of the He-Xe gas, and M_{mix} is the molar mass of the He-Xe gas. The calculation formula is as follows:

$$\rho_{mix} = \frac{P_{mix}}{(R_g T Z)} \cdot M_{mix} = \frac{P_{mix} \cdot M_{mix}}{R_g T(\hat{\rho} + \overline{B}\hat{\rho}^2 + \overline{C}\hat{\rho}^3)} \tag{39}$$

$$M_{mix} = x_1 \cdot M_1 + x_2 \cdot M_2 \tag{40}$$

where P_{mix} is the pressure exerted on the He-Xe gas, Z is the compressibility factor, x_1 and x_2 represent the mole fractions of He and Xe gases in the He-Xe mixture, and M_1 and M_2 are the molar masses of He and Xe gases, respectively. The molar heat capacity at constant pressure and the ratio of specific heats of He-Xe gas under the conditions of 1000K and 2 MPa vary with molar mass, as shown in Fig. 12. Taking a mixture of 72% He and 28% Xe as an example [41], the density of He-Xe gas as a function of temperature is shown in Fig. 13, and the relationship between C_P and C_V is shown in Fig. 14. For a detailed calculation of the Virial $\overline{B}$ and $\overline{C}$ coefficients, refer to Appendix A.1.

The convective heat transfer coefficient h of the working fluid can be calculated according to formulas (33) and (34) in Section 2.3:

$$h = \frac{Nu \cdot \lambda}{D} \tag{41}$$

$$Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{0.4} \tag{42}$$

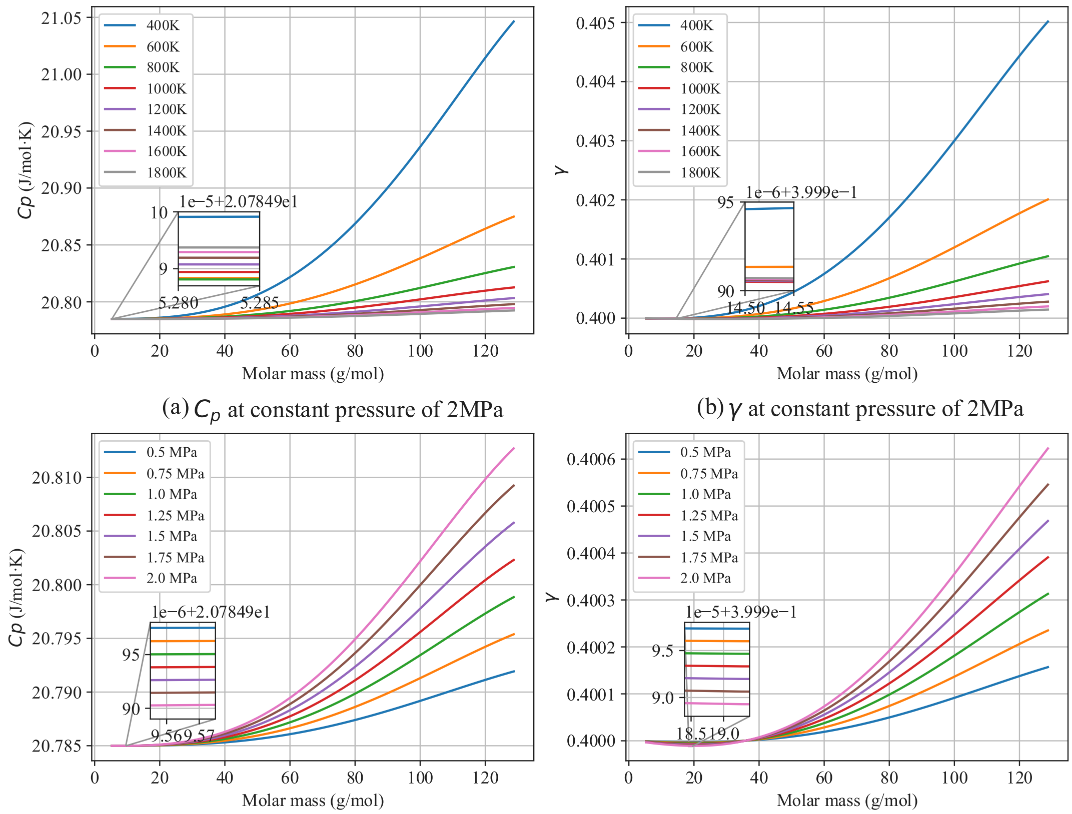
$$Re = \frac{\rho_{mix} \cdot u \cdot D}{\mu} \tag{43}$$

$$Pr = \hat{C}_p\frac{\mu}{\lambda} \tag{44}$$

where Nu is the Nusselt number, λ is the thermal conductivity, Re is the Reynolds number, Pr is the Prandtl number, u is the velocity, μ is the dynamic viscosity, and D is the hydraulic diameter. Both dynamic viscosity and thermal conductivity must be determined using semi-empirical formulas [58–60], see Appendices A.2 and A.3 for detailed calculation. The relationship between Prandtl number and molar mass is shown in Fig. 15, while the variations of dynamic viscosity and thermal conductivity with temperature are shown in Fig. 16.

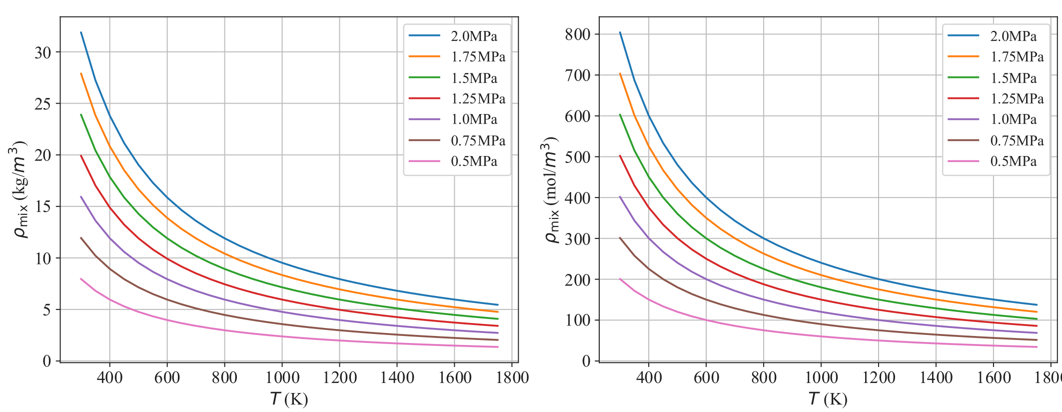
The physical properties of the metals constituting each component, as well as the properties of the NaK and Li working fluids [52,61], are illustrated in Fig. 17. The corresponding formulas can be found in Appendix A.4.

Once the models for each component are established, parameters and initial values can be specified individually to verify and investigate local characteristics. Alternatively, as depicted in Fig. 18, the components can be coupled to study the overall characteristics of the SNRCBC



(a) C_p at constant pressure of 2MPa
(b) γ at constant pressure of 2MPa
(c) C_p at constant pressure of 1000K
(d) γ at constant pressure of 1000K

Fig. 12. Molar Specific Heat and Heat Capacity Ratio of He-Xe Gas vs. Molar Mass.



(a) Mass density
(b) Molar density

Fig. 13. Density of He-Xe Gas (72%He & 28%Xe) vs. Temperature.

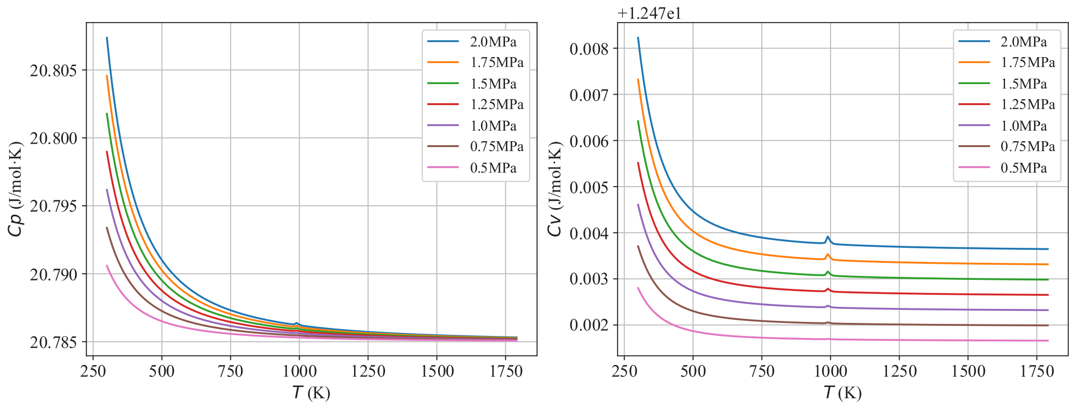
system. In this context, the solution of the differential equations within the model is approached using the finite difference method (FDM), while numerical values requiring iterative solutions in the heat transfer calculations are addressed through the Newton's method and adaptive gradient descent method [62–64].

3. Simulation results and analysis

In this section, we first establish the key parameters of the experiment, neglecting the pressure losses in the pipelines connecting the various components, and employ a single-stage coupled TAC power generation. A one-dimensional model of the SNRCBC was constructed by integrating Python, Refprop, and simulation software. The steady-state simulation was calibrated, and transient simulations of the startup process were conducted, thereby confirming the accuracy and correctness of both the individual component models and the overall system model. Four typical transient operating conditions, including varying shaft speed, varying reactivity, varying working fluid flow, and varying load, are simulated to analyze their characteristics. The models of the components and coupled systems are shown in Fig. 19, the design values and parameters are referenced from literature [48] and report [53].

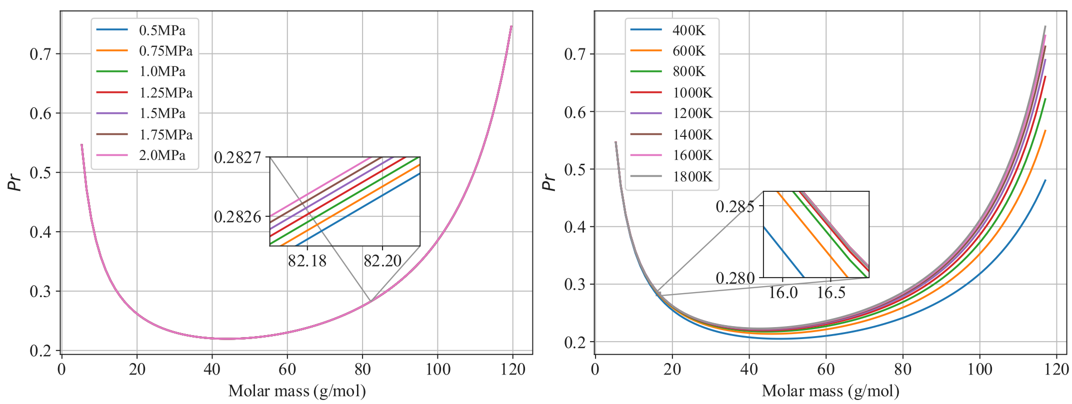
3.1. The steady-state operating condition

The static parameters are presented in Table 3 (It is worth noting that aluminum alloy is used for the radiative heat sink to reduce weight



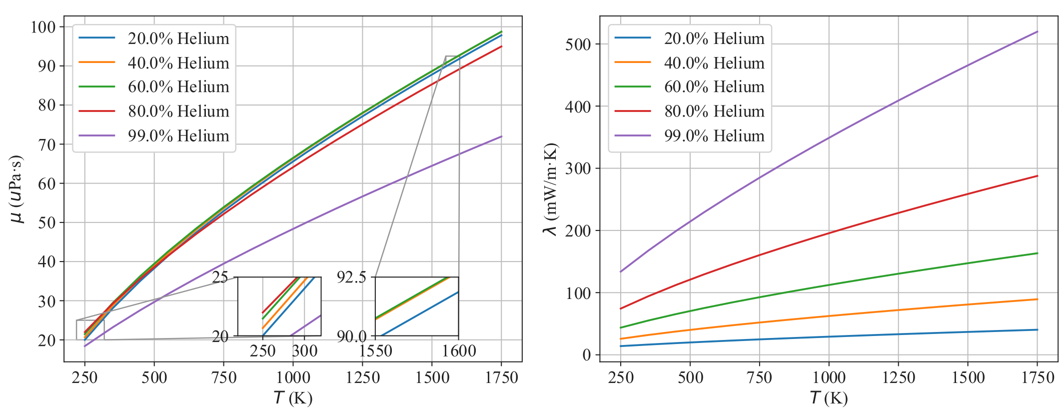
(a) Molar heat capacity at constant pressure (b) Molar heat capacity at constant volume

Fig. 14. Molar heat capacity of He-Xe Gas (72%He & 28%Xe) vs. Temperature.



(a) Pr at constant pressure of 1000K (b) Pr at constant pressure of 2MPa

Fig. 15. Prandtl number of He-Xe Gas vs. Temperature.



(a) Variation of Dynamic viscosity (b) Variation of Thermal conductivity

Fig. 16. Dynamic viscosity and thermal conductivity of He-Xe Gas vs. Temperature.

in practical applications. Although its emissivity is low, surface coating can raise it to 0.9 without compromising the accuracy of the internal heat transfer model, all the data is referenced from Report [53]) and studies [48,65]. To ensure optimal performance under the strict volume and weight constraints of space reactors, careful selection of the working fluid is essential. Prior studies [38,41] indicate that a helium–xenon mixture with a molar mass around 40 g/mol offers a suitable balance. Helium provides excellent thermal conductivity but increases system mass due to its low molecular weight and resulting high specific volume. In contrast, xenon's higher molecular weight mitigates these drawbacks by reducing the total volume and piping requirements. Based on this trade-off, the present study adopts a mixture of 72% helium and 28% xenon. The reactor loop utilizes liquid lithium, while the coolant in the radiator is Nak alloy (22Na-78K), consisting of 22% sodium (Na) and 78% potassium (K). The freezing or melting process of the metal coolant is not considered, and the space temperature is

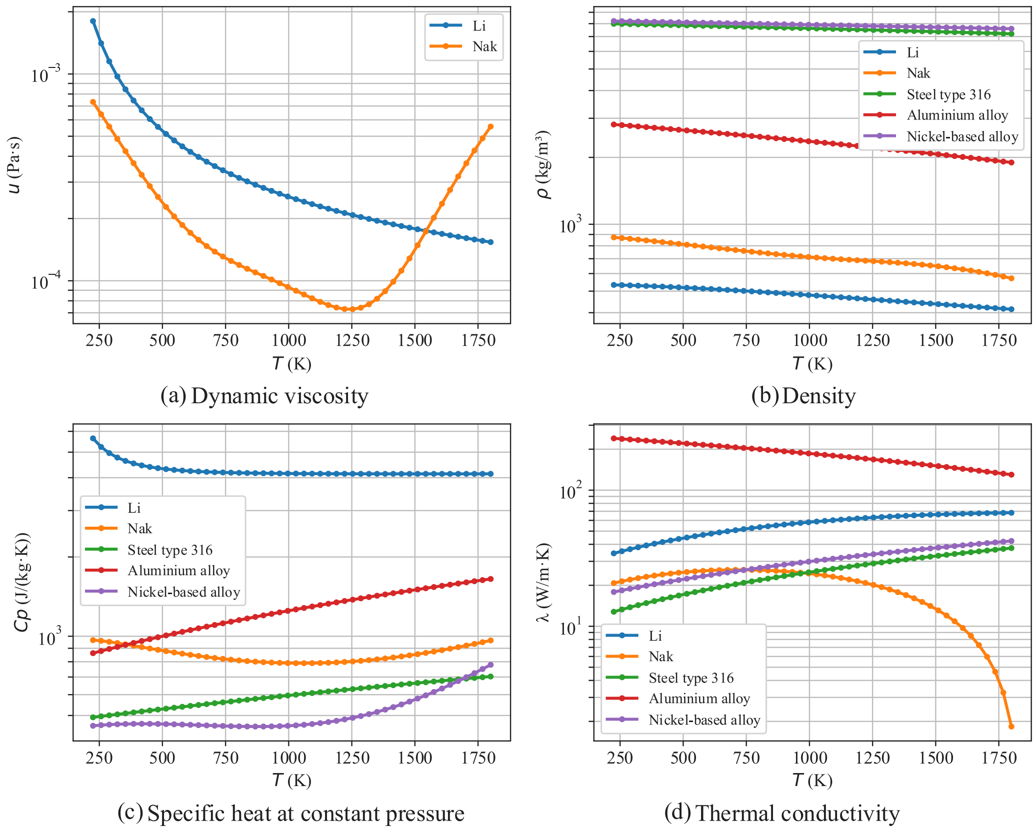


Fig. 17. Properties of the metals vs. Temperature.

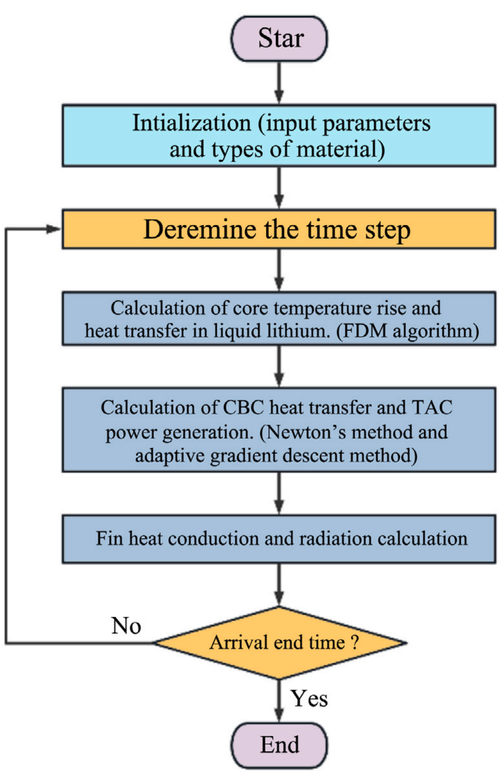


Fig. 18. Flow chart of the program.

set to 225 K. During steady-state operation, the total reactivity in the reactor is 0, and the TAC shaft speed is maintained at a constant value. The simulation results are depicted in Figs. 20 to 23.

Initially, maintaining a constant inlet temperature of 225 K and the reactivity insertion is 0.009$ (reactivity is a dimensionless parameter. The explanation of the unit $ can be found in the Refs. [66].). Liquid lithium is employed as the coolant, with a constant mass flow rate of 4.572 kg/s. The initial neutron density is set to $1\mathrm{n/m^3}$, and the initial concentration of delayed neutron precursors is also set to $1\mathrm{mol/m^3}$. The variations in both the core temperature and outlet temperature are shown in Fig. 20. It is evident that the core temperature increases at a higher rate than the outlet temperature, and the steady-state core temperature is also greater than the outlet temperature. This indicates that, while the temperature of the liquid lithium coolant remains constant at 225 K at the inlet, it absorbs part of the heat generated by the core as it passes through, resulting in a rise in the outlet temperature, though it does not exceed the core temperature. This provides preliminary validation of the accuracy of the core model.

As shown in Fig. 21, the temperature variation process of the four heat exchanger components before reaching steady state is shown. The input parameters for each component module correspond to the steady-state values. It can be observed that both the outlet temperature and the component temperature of each unit exhibit a pattern of rapid initial increase followed by a gradual slowdown as they approach a stable value. The component temperature rises the fastest initially, followed by the cold-end outlet temperature, and finally the hot-end outlet temperature. A notable difference is that the time at which the temperature rise starts for each hot-end outlet varies, as does the time it takes to reach steady state. Among them, the heater has the fastest overall response, stabilizing after approximately 30 s, followed by the radiator at around 90 s, and the recuperator and cooler at about 100 s. Except for the cooler, whose temperatures at both ends respond immediately at time zero, the heater's outlet temperature begins to rise

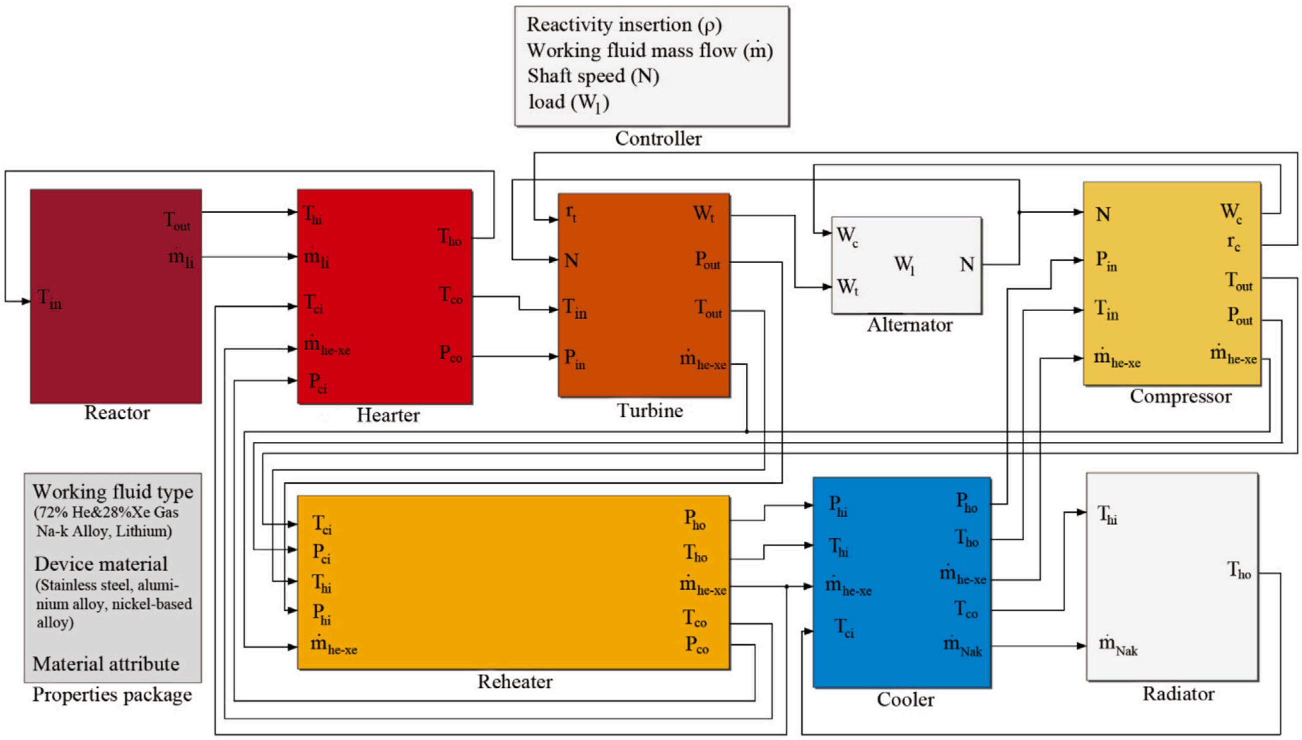


Fig. 19. Model of He-Xe Brayton Cycle Power Generation System for Megawatt-Scale Space Lithium-Cooled Reactor.

Table 3
Static parameters.

Parameter	Reactor core	Heater	Recuperator	Cooler	Radiator
Core fuel volume (V_f)	1 m^3	–	–	–	–
Fission energy release (E_f)	200 MeV	–	–	–	–
Fission cross-section (Σ_f)	0.1 cm^{-1}	–	–	–	–
Length (l)	–	5 m	4 m	4 m	6 m
Equivalent diameter (D)	–	0.008 m	0.017 m	0.014 m	0.019 m
Wall thickness (r)	–	0.002 m	0.002 m	0.002 m	0.004 m
Number of pipes	–	50	50	50	50
Material	Uranium-235	Nickel-based alloy	Nickel-based alloy	Stainless steel	Aluminum alloy
Emissivity (ϵ)	–	–	–	–	0.9
Heat dissipation area(A_{rad})	–	–	–	–	232 m^2
Flow channel volume	–	0.001 m^3	0.00363 m^3	0.00246 m^3	0.00721 m^3

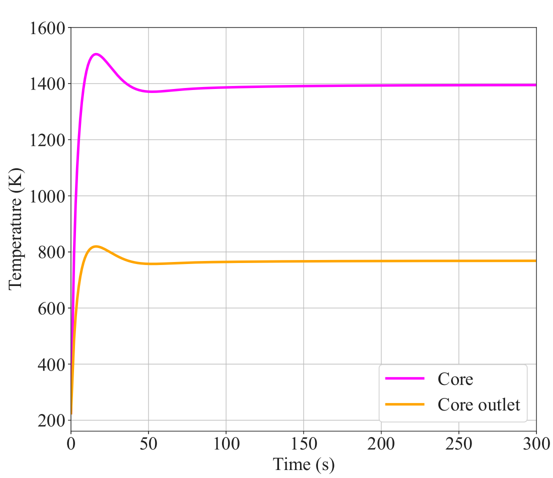


Fig. 20. Core and Core outlet temperature.

after 3 s, while the recuperator and radiator start rising at the 6th and 12th seconds, respectively. The differences in the rate of temperature change between components are mainly attributed to variations in the internal structure, heat transfer coefficients, and other parameters of the heat exchangers.

The outlet temperatures of the turbine and compressor are shown in Fig. 22. Since the inlet temperature of the turbine is identical to the cold-end outlet temperature of the heater, the helium-xenon coolant is heated to a relatively high temperature after passing through the heater. Even after expansion in the turbine, the temperature remains higher than the compressor's outlet temperature. Similarly, the compressor's inlet temperature is the same as the hot-end outlet temperature of the cooler. After the helium-xenon coolant is cooled by the cooler, its temperature decreases, and even after compression in the compressor, the temperature rise will not exceed that of the turbine's outlet temperature.

Fig. 23 illustrates the power variation process of the system's main components before reaching steady state, including the reactor power, turbine power, compressor power, and power generated for load. The results show that the power of the main components initially undergoes dynamic changes, stabilizing after approximately 100 s. Once steady state is reached, the reactor core power is approximately 3.248MW, the turbine power is 2.849MW, the compressor power is 1.777MW, and the TAC power generation is 1.072MW. At steady state, the shaft speed corresponding to the TAC shaft is 55,000rpm.

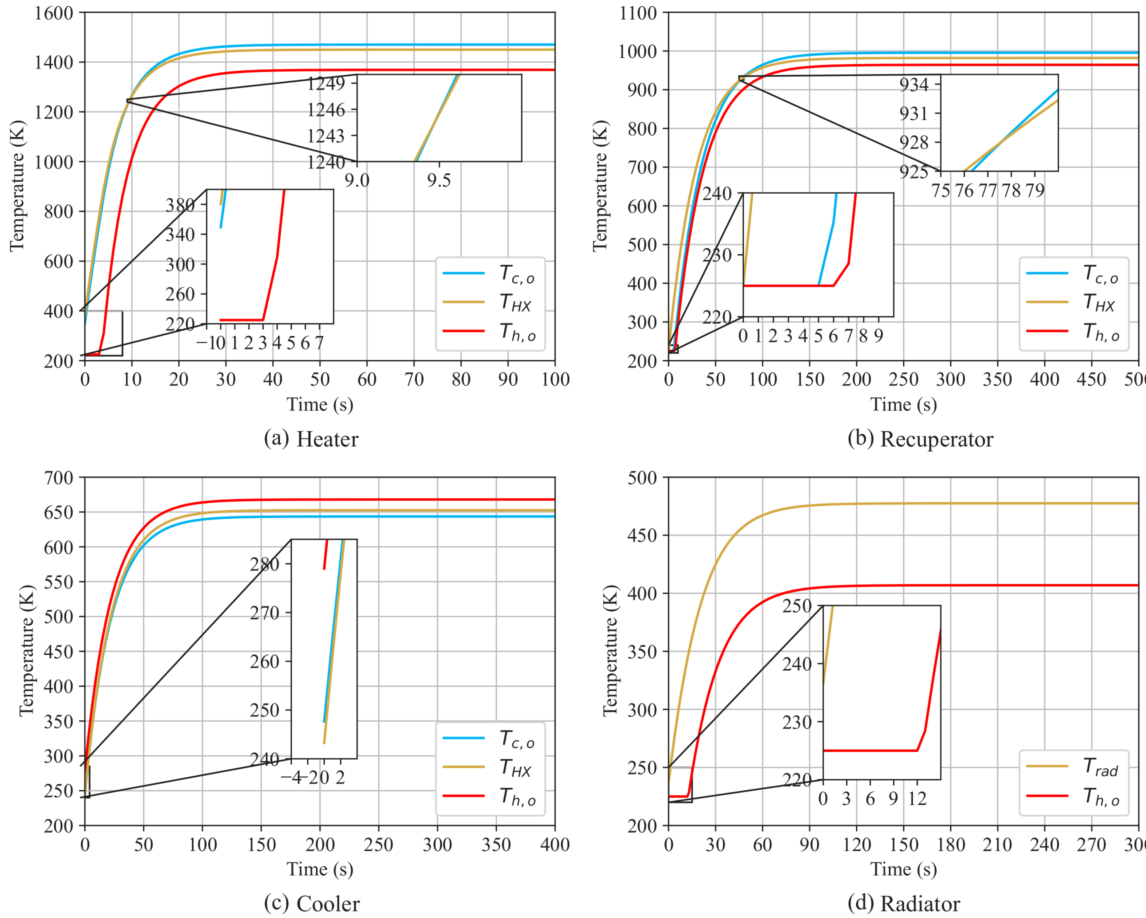


Fig. 21. Outlet temperatures and structural temperature of heat exchangers under steady-state conditions.

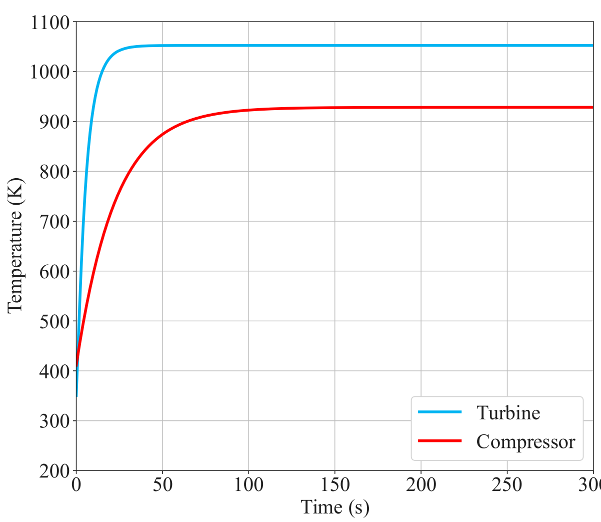


Fig. 22. Outlet temperatures of Turbine and compressor.

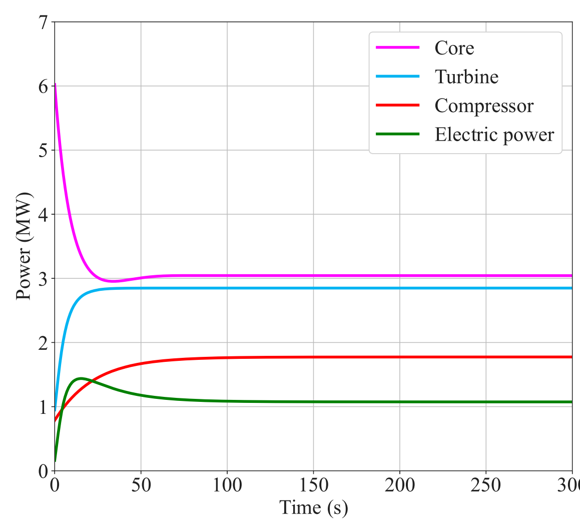
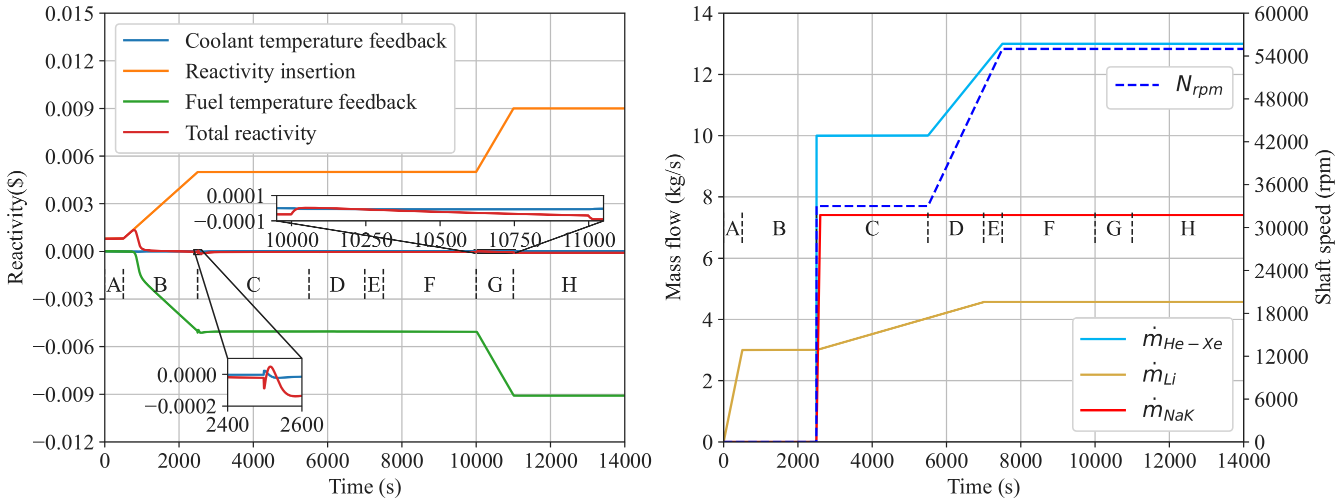


Fig. 23. Reactor core and TAC power under steady-state conditions.

3.2. The transient startup process

The system startup plan is designed while keeping the static parameters unchanged. The dynamic parameters during the startup process are illustrated in Fig. 24. There are generally two startup methods for the SNRCBC: one is to start the TAC first, followed by the core, and the other is to start the core first, followed by the TAC. We adopt the second method, as starting the core before the TAC reduces the demand for external auxiliary motors.

The startup process is divided into eight sequential stages, labeled A through H. This scheme draws upon the startup procedures proposed in Refs. [40,48] for small-scale Brayton systems and is governed by a rule-based control module. In this control logic, the transition to each subsequent stage occurs only when specific temporal and physical conditions are met, with detailed operational criteria described within each



(a) Core reactivity (b) Working fluid flow and Shaft speed

Fig. 24. Dynamic parameters.

corresponding phase. The rationale for this approach lies in practical engineering constraints. To minimize the impact of liquid sloshing and associated dynamic instabilities during launch, the reactor coolant is initially configured as solid-state metallic lithium. Therefore, in the post-launch space environment, the reactor startup must begin with a thawing phase to liquefy the coolant before activating the Brayton cycle system. Once the coolant reaches a sufficiently high temperature, the power conversion system is gradually engaged. The startup sequence is designed to enhance operational safety and includes preheating the reactor core, initiating the Brayton cycle at low temperature and low power to absorb heat, increasing to full shaft speed while maintaining low power conditions, and finally reaching steady-state operation at full power. Due to the complexity of the heat transfer mechanisms involved in the thawing process, and because this is not the primary focus of the present study this phase is modeled by the time-varying mass flow rate of metallic lithium, as shown in 24(b).

Phase A: This phase marks the reactor power ramp-up phase, during which the reactor needs to increase its initial power to a kilowatt-level range, thereby generating latent heat effects in preparation for subsequent thermal feedback and the rise in reactor temperature. The initial reactor power is set to 1 W, and the mass flow rate of the liquid lithium coolant within the core increases linearly to 3 kg/s. The initial reactivity insertion is 0.0008$, and as the core power increases, it reaches 1.01kW approximately 500 s later, as shown in the A interval of Fig. 25.

Phase B: This phase corresponds to the reactor core temperature rise stage, during which the reactor power and core temperature are increased by controlling the magnitude of reactivity inserted into the core. To prevent the reactor temperature from rising too rapidly, the reactivity is linearly increased from 0.0008$ to 0.005$ over a period of 2000 s, as shown in Fig. 26. At this point, the core temperature reaches 710 K, and the core power reaches 1.562MW. Throughout this stage, the mass flow rate of the liquid lithium coolant remains constant. It is important to note that this paper considers the heater's temperature rise due to heat conduction from liquid lithium during this stage (not convective heat transfer), as shown in the B interval of Fig. 27 (a). This ensures that the heater's temperature increase is accounted for when the Brayton cycle starts, improving model accuracy. Existing studies have overlooked this and assumed a fixed heater temperature at the start of the cycle. [38–40,48].

Phase C: This phase corresponds to the startup stage of the Brayton cycle. The TAC shaft in the Brayton cycle power generation system begins to rotate, driving the turbine and compressor. The shaft speed

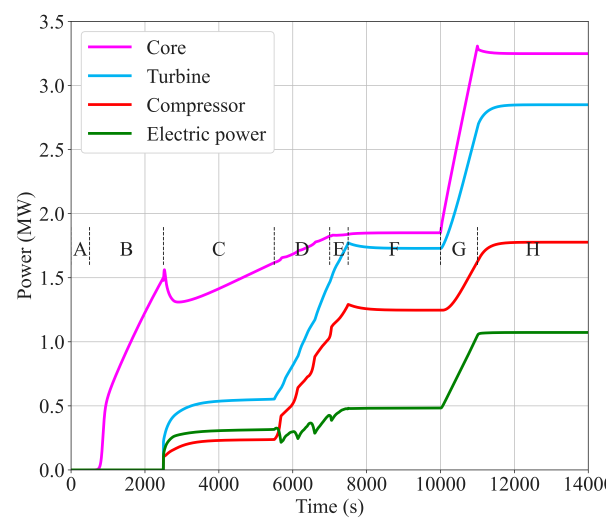


Fig. 25. Reactor core and TAC power.

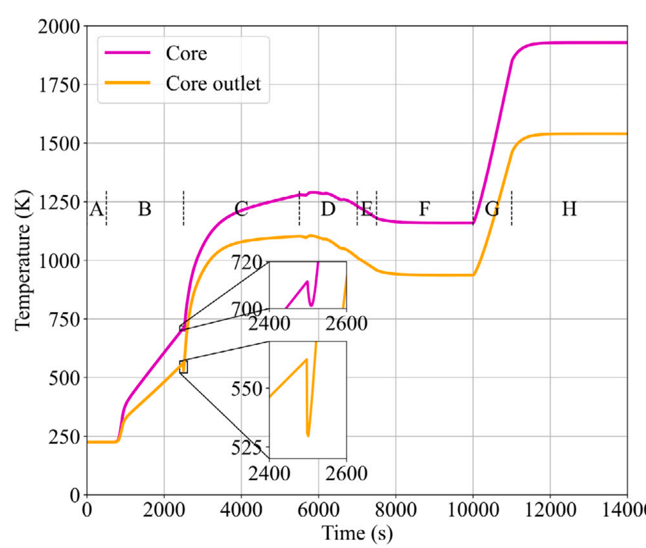


Fig. 26. Core and outlet temperatures.

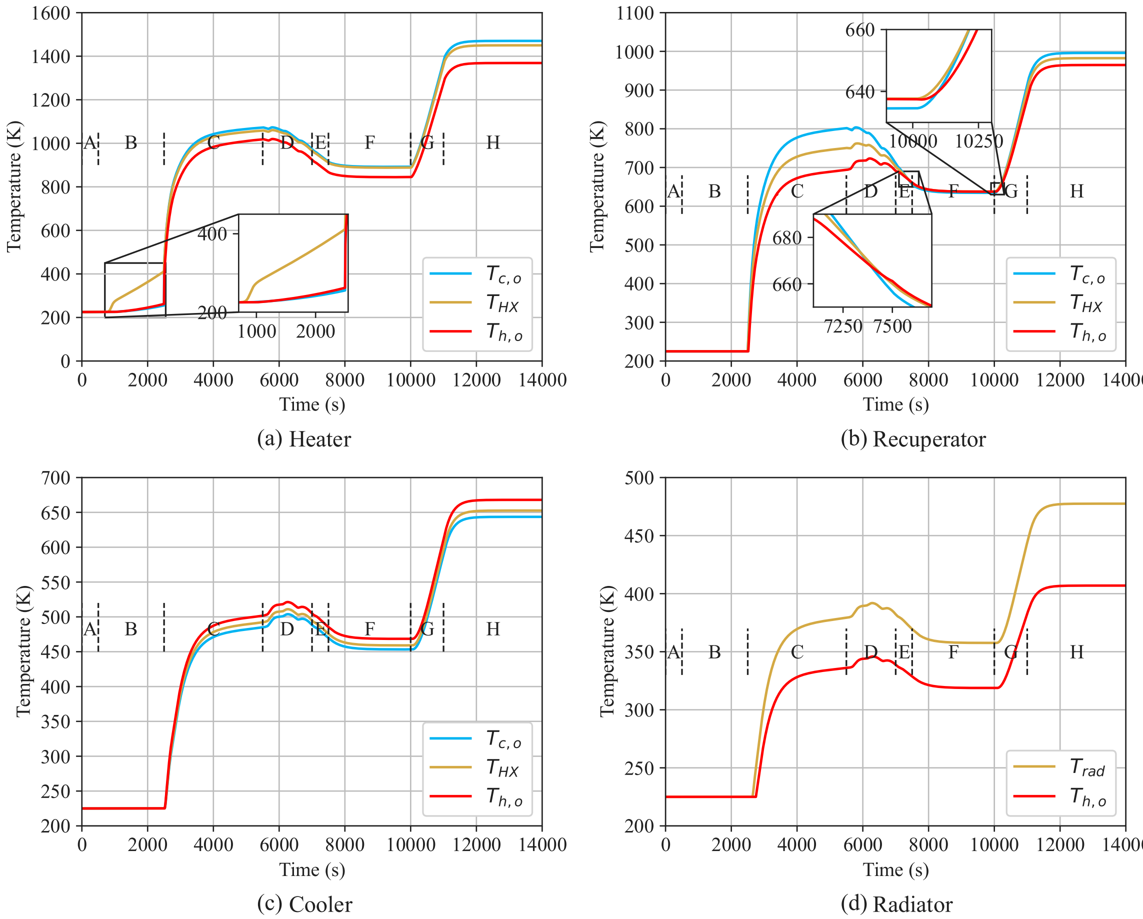


Fig. 27. Outlet temperatures and structural temperature of heat exchangers during startup.

is controlled to reach 33,000rpm, with a He-Xe gas mass flow rate of approximately 10 kg/s. Subsequently, the inlet and outlet temperatures of various components rise rapidly, and the radiator also begins to operate. The mass flow rate of the NaK coolant is approximately 7.408 kg/s, as shown in Fig. 24(b). Meanwhile, the mass flow rate of liquid lithium increases linearly. After a brief period, the power of the turbine and compressor stabilizes at 0.554MW and 0.237MW, respectively, causing the power used for load generation to cease increasing. The inlet and outlet temperatures of each component also stabilize, as shown in Fig. 27. During this phase, the power generation and core power levels are relatively low, making it an ideal time for inspection and diagnostics of the system before the operation of various devices, facilitating the detection of any issues within the system. As a result, the design operating time is relatively long, reaching 3000 s.

Phase D: This phase is the TAC acceleration stage, with a duration of 1500 s. The TAC shaft speed rises linearly, and the mass flow rate of He-Xe gas also increases proportionally. The mass flow rate of liquid lithium reaches 4.57 kg/s. During this time, the power of both the turbine and compressor rises, while the reactivity insertion remains constant from stage C onwards. Due to thermal feedback effects, the total reactivity approaches zero, and the power increase gradually slows down.

Phase E: This phase marks the final acceleration stage of the TAC, lasting 500 s. The TAC shaft speed reaches 55,000rpm, while the He-Xe gas mass flow rate also attains 13 kg/s. The core power increases to 1.837 MW, with the turbine and compressor powers rising to 1.772MW and 1.291MW, respectively.

Phase F: This phase represents the first steady-state operation period, during which the shaft speed, He-Xe gas mass flow rate, and liquid lithium mass flow rate all reach their design values. The core power, turbine power, and compressor power stabilize and no longer change, with the inlet and outlet temperatures of each component also reaching stability. The system maintains a steady state for 2,000 s with its overall parameters unchanged. According to the characteristic curves of the turbine and compressor, since the He-Xe mass flow rate and shaft speed remain constant, the inlet and outlet pressure ratios of both the turbine and compressor are fixed, as shown in Fig. 28.

Phase G: This phase marks the full-power operation stage of the reactor. During this period, reactivity is linearly increased to 0.009$, causing both the core power and temperature to rise accordingly, until the power generated by the TAC for load meets the design requirement at the MW level. Ultimately, the core power stabilizes at 3.248MW, the turbine power at 2.849MW, the compressor power at 1.777MW, and the power generated for load reaches 1.072MW, the power generation efficiency is 33.03%.

Phase H: This phase represents the final stabilization stage, with the SNRCBC system fully activated. Unlike what is depicted in Fig. 20, the core inlet temperature varies in accordance with the changes in the heater's hot-end outlet temperature, yet the maximum temperature does not exceed 2000 K. The temperatures of other components, as shown in Fig. 27, remain within safe limits and adhere to the principles of heat transfer, thereby validating the accuracy of the transient model. The corresponding design values are presented in Table 4 and are compared with the existing studies as illustrated in Fig. 29.

The simulation results indicate that the proposed approach offers clear advantages in terms of parameter completeness and system-level performance. Numerically, all simulated values fall within physically reasonable and technically meaningful ranges. The reactor core outlet temperature reaches 1539.68 K, significantly higher than the 1200 K and 1070 K reported in Ref. [39] and Report [53], respectively, and

Table 4
Design values and simulation values.

Parameter	Design value	Simulated value	Relative error (±%)	Ref [38]	Ref [39]	Ref [41]	Report [53]
Reactor core outlet temperature	1540 K	1539.68 K	0.02%	1500 K	1200 K	1150 K	1070 K
Heater cold side inlet temperature	996 K	996.64 K	0.06%	No part	854 K	1099 K	No part
Heater cold side outlet temperature	1470 K	1468.29 K	0.11%	No part	1153 K	1150 K	No part
Turbine outlet temperature	1055 K	1052.19 K	0.26%	1123 K	915 K	1163 K	852 K
Turbine inlet pressure	2.05 MPa	2.04 MPa	0.16%	3.30 MPa	2.96 MPa	0.8~1 MPa	2.56 MPa
Turbine outlet pressure	0.75 MPa	0.76 MPa	1.4%	1.38 MPa	1.51 MPa	0.6~0.8 MPa	2.4 MPa
Recuperator hot side outlet temperature	960 K	964.76 K	0.49%	713 K	568 K	Not specified	574 K
Cooler hot side outlet temperature	670 K	667.95 K	0.3%	No part	362 K	Not specified	No part
Cooler cold side outlet temperature	640 K	643.58 K	0.55%	No part	463 K	Not specified	No part
Radiator hot side outlet temperature	405 K	406.94 K	0.47%	405 K	345 K	Not specified	395 K
Compressor outlet temperature	930 K	928.63 K	0.14%	640K	507 K	Not specified	545 K
Compressor inlet pressure	0.75 MPa	0.76 MPa	1.4%	1.38 MPa	1.5 MPa	0.6 ~0.8 MPa	2.4 MPa
Compressor outlet pressure	2.05 MPa	2.04 MPa	0.16%	3.30 MPa	3.0 MPa	0.8 ~1 MPa	2.56 MPa
Generated power	1 MW	1.072 MW	0.07%	273 kW	161 kW	150.82 kW	100 kW
Efficiency	33.33%	33.03%	0.9%	32.10%	31.10%	20.71%	22.90%

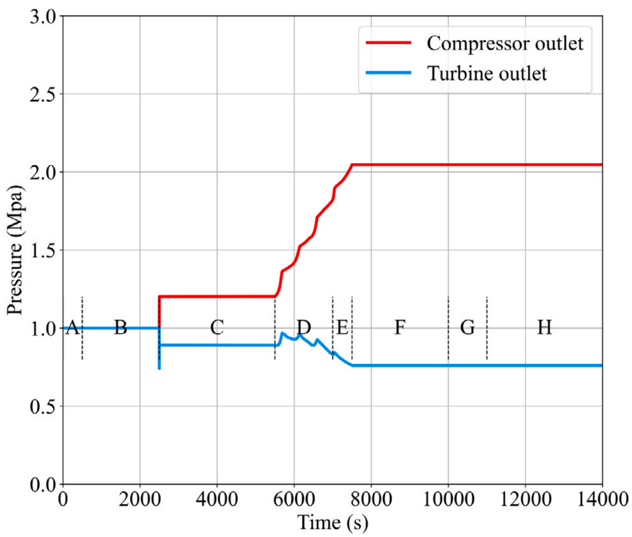


Fig. 28. Outlet pressure of the turbine and compressor.

slightly above the 1500 K in Ref. [38], aligning more closely with the expectations of high-performance gas-cooled reactor concepts. Similar consistency is observed across other critical components. The simulated turbine outlet temperature of 1052.19 K provides a thermally efficient expansion endpoint, while the compressor outlet temperature of 928.63 K indicates sufficient compression work. In contrast, the significantly lower values reported in Ref. [39] (507 K) and Report [53] (545 K) may result from simplified assumptions or conservative design constraints. In terms of pressure characteristics, the simulated turbine inlet and outlet pressures are 2.04 MPa and 0.764 MPa, respectively. These values reflect a practical and balanced pressure ratio suitable for closed Brayton cycle applications in space. By comparison, Ref. [39] reports higher values of 2.96 MPa and 1.51 MPa, while Ref. [41] adopts a lower-pressure regime of 0.8 MPa and 0.6 MPa. The simulated pressures in this study lie within a moderate range, indicating a trade-off between thermodynamic efficiency and mechanical simplicity. With respect to overall performance, the simulated net power output reaches 1.072MW with a thermal efficiency of 33.03%, clearly exceeding the reported values of 0.273MW (32.1%) in Ref. [38], 0.161MW (31.1%) in Ref. [39], 0.15MW (20.71%) in Ref. [41], and 0.1MW (22.9%) in Report [53]. Moreover, the partial data omissions and simplified component representations in Ref. [38], Ref. [41], and Report [53] limit the completeness of system-level comparisons.

These improvements highlight the effectiveness of thermal integration and component matching achieved in the present model. Collectively, the simulated results not only provide a more complete and physically consistent dataset, while affirming the proposed steady-state model's scientific rigor and engineering applicability in thermophysical coupling, parameter tuning, and system integration.

3.3. Variable condition simulation

After the experimental verification of the SNRCBC component model and the overall system model, it is also an indispensable research content for the analysis of the operation characteristics of the simulation system under varying operating conditions during the dynamic operation. This section primarily examines four operational conditions: variations in reactivity, variable shaft speed, fluctuating working fluid flow, and changing load, all of which focus on the post-steady-state phase following the startup process in the Phase H of Section 3.2.

Condition 1 (*Varying Reactivity*)**.** After operating under steady-state conditions for 1000 s, a positive reactivity step of +0.003$ is applied for 2000 s, followed by a negative reactivity step of −0.003$ for another 2000 s, restoring the reactivity to its initial value of 0.009$ as defined in the startup plan. Subsequently, a reactivity reduction of −0.004$ is introduced, decreasing the reactivity from 0.009$ to 0.005$. After a further 2000 s of operation, a +0.005$ reactivity increment is applied, elevating the total reactivity to 0.01$, as illustrated in Fig. 30

Whenever reactivity changes, the reactor power responds swiftly. Upon the application of a +0.005$ reactivity increment, the system's total reactivity increases instantaneously, resulting in a sharp rise in reactor power to 6.391 MW, as shown in interval Δ_1 of Fig. 31(a). However, under the moderating effect of the negative feedback from the core temperature, the reactivity diminishes, leading to a subsequent reduction in reactor power and eventually stabilizing at 4.299 MW. Due to the rise in reactivity and core power, the core outlet temperature also increases, thereby enhancing the electrical output of the TAC system, ultimately reaching 1.564 MW, as shown in Fig. 31(b). When a −0.003$ reactivity step is applied, the core power swiftly drops to 2.472 MW. Similarly, due to the temperature-based negative feedback, the power gradually stabilizes back to 3.248 MW, with the TAC output returning to 1.072 MW. Following this, a further −0.004$ reactivity decrement reduces the reactivity to 0.005$, lowering the core power to 1.850 MW and the TAC output to 0.578 MW. Finally, a +0.005$ reactivity increment brings the reactivity back up to 0.01$, resulting in a rise in core and TAC power to 3.599 MW and 1.124 MW, respectively. The results clearly demonstrate a consistent correlation between variations in introduced reactivity and the corresponding fluctuations in core power and TAC electrical output, aligning well with theoretical expectations and thereby validating the model's accuracy. Furthermore, the presence of the temperature-based negative feedback mechanism ensures that abrupt changes in core power quickly stabilize, further confirming the model's reliability.

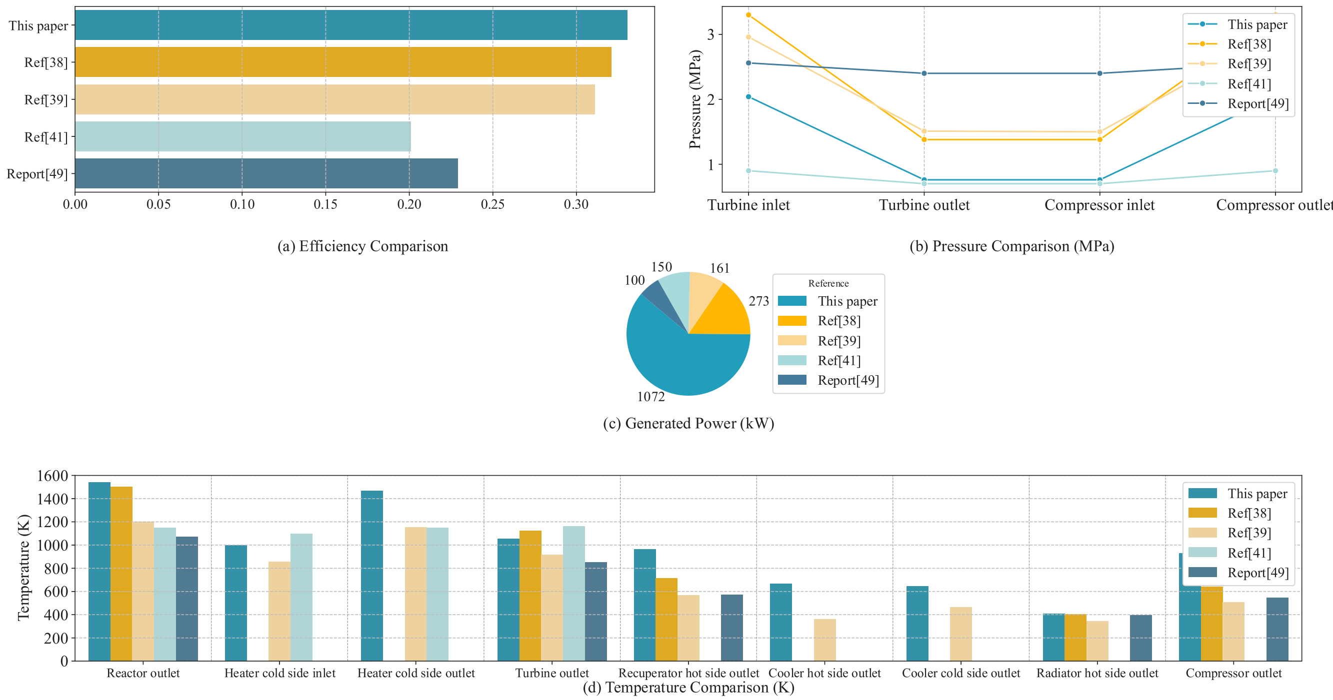


Fig. 29. Compared with the existing studies.

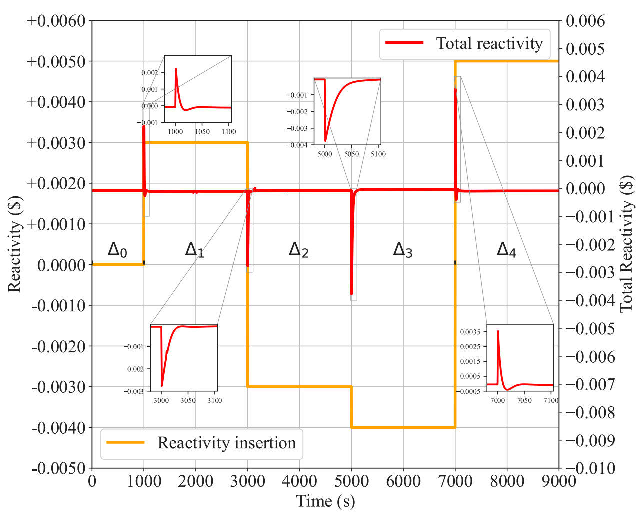
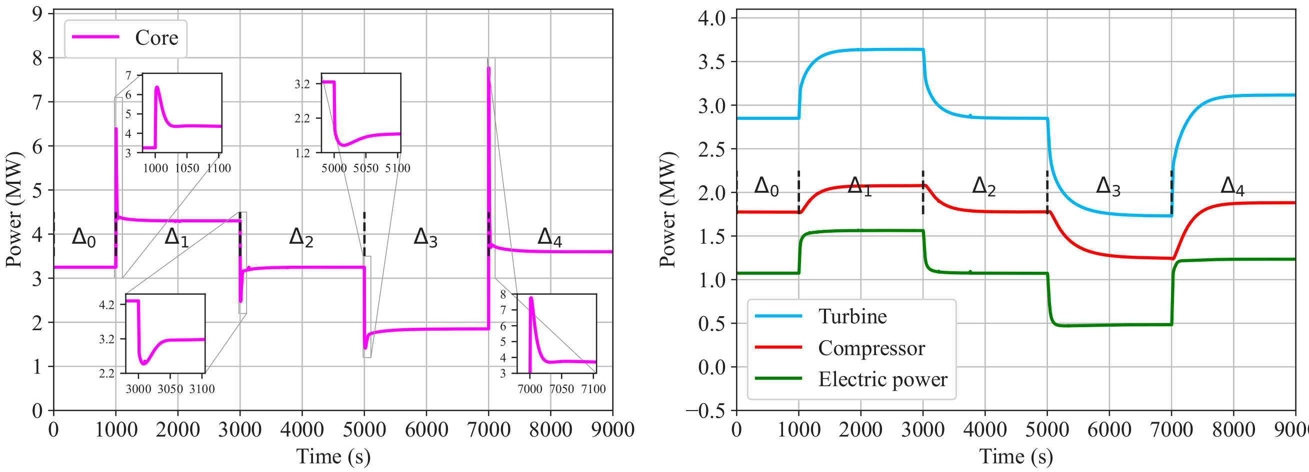


Fig. 30. Variations in reactivity.

Condition 2 (*Variable Shaft Speed*)**.** The variation in TAC shaft speed is set as shown in Fig. 32. The system maintains a shaft speed of 55,000 rpm for 1000 s, after which it abruptly drops to 47,000 rpm. Following 2000 s of operation at this speed, it further decreases to 35,000 rpm. After another 2000 s, the speed rises to 36,000 rpm, and finally returned to 55,000 rpm after another 2000 s.

The corresponding changes in TAC power are shown in Fig. 33(a). It can be seen that as the shaft speed decreases from 55,000 rpm to 47,000 rpm during phase Δ_1, and remains at 35,000 rpm in phase Δ_2, the power outputs of both the turbine and the compressor decrease. Conversely, as the shaft speed increases from 35,000 rpm to 36,000 rpm in phase Δ_3, and further increases to return to 55,000 rpm in phase Δ_4, the power outputs of both components increase accordingly. However, what is different is that the generator power output increases during phases Δ_1, Δ_2, and Δ_3, and only decreases in phase Δ_4. This is because the shaft speed affects the pressure ratios of the turbine and compressor, which in turn influence the temperature differences between their inlets and outlets, i.e., the corresponding enthalpy changes. When the enthalpy rise in the compressor exceeds the enthalpy drop in the turbine, the compressor's power variation dominates. For example, if both power outputs decrease simultaneously, and the compressor power decreases more, the generator power will increase, as shown in Eqs. (17) and (20), TAC Power $= W_T - W_C$). As shown in Fig. 34(a) (see Figs. 8 and 9 for the full characteristic curves), when the speed increases from 35,000 rpm to 36,000 rpm, the turbine pressure ratio increases by +0.036, while the compressor pressure ratio increases by only +0.029. As a result, during phase Δ_3, the enthalpy rise of the compressor is not as significantly larger than the enthalpy drop of the turbine as in phase Δ_2, as shown in Fig. 34(b) (The figure shows the absolute values of the temperature changes at the inlet and outlet of the turbine and compressor, as $|T_{out} - T_{in}| - |T'_{out} - T'_{in}|$, T' represents the inlet and outlet temperatures of the turbine or compressor under steady-state conditions), leading to an increase in generator power.

As shown in Fig. 33(b), the core power remained stable at 3.248 MW during the first 1000 s. At 1000 s, when the shaft speed dropped from 55,000 rpm to 47,000 rpm, the core power increased sharply, then decreased rapidly, followed by another increase, and finally stabilized at 3.2536 MW. At 3000 s, the speed decreased further to 35,000 rpm, causing the core power to rise quickly, then drop, and then rise again before stabilizing at 3.5226 MW. At 5000 s, the speed increased to 36,000 rpm, and the core power first dropped rapidly, then rose to 3.2531 MW. Finally, at 7000 s, when the speed returned to 55,000 rpm, the core power decreased sharply, then rose briefly, and finally decreased again to stabilize at 3.248 MW. This behavior is due to the sudden change in shaft speed while maintaining a constant He-Xe coolant mass flow rate. When the speed drops, the outlet temperatures of both the turbine and compressor decrease. As a result, the He-Xe coolant removes more heat from the core, leading to a drop in core temperature. Due to the reactivity feedback mechanism, this enhances positive reactivity, causing the core power to increase temporarily. As the system temperature rises quickly thereafter, it triggers a negative feedback, reducing reactivity and bringing the power back to a stable value after a brief fluctuation. Conversely, when the speed increases,



(a) Core power (b) TAC power

Fig. 31. Power variations under changes in reactivity.

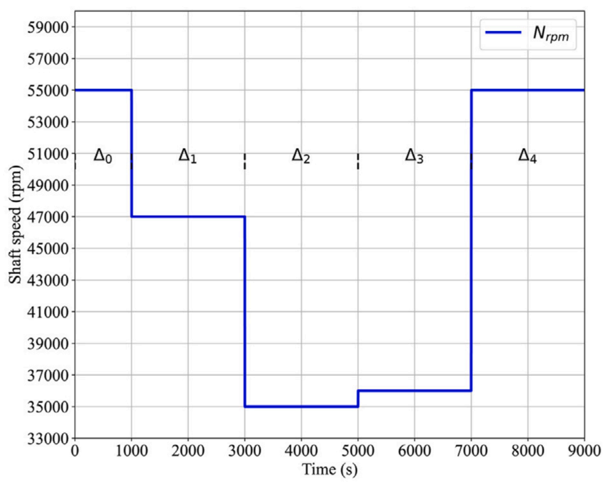


Fig. 32. Variations in shaft speed.

the core power initially decreases and then rises, eventually stabilizing after a transient fluctuation.

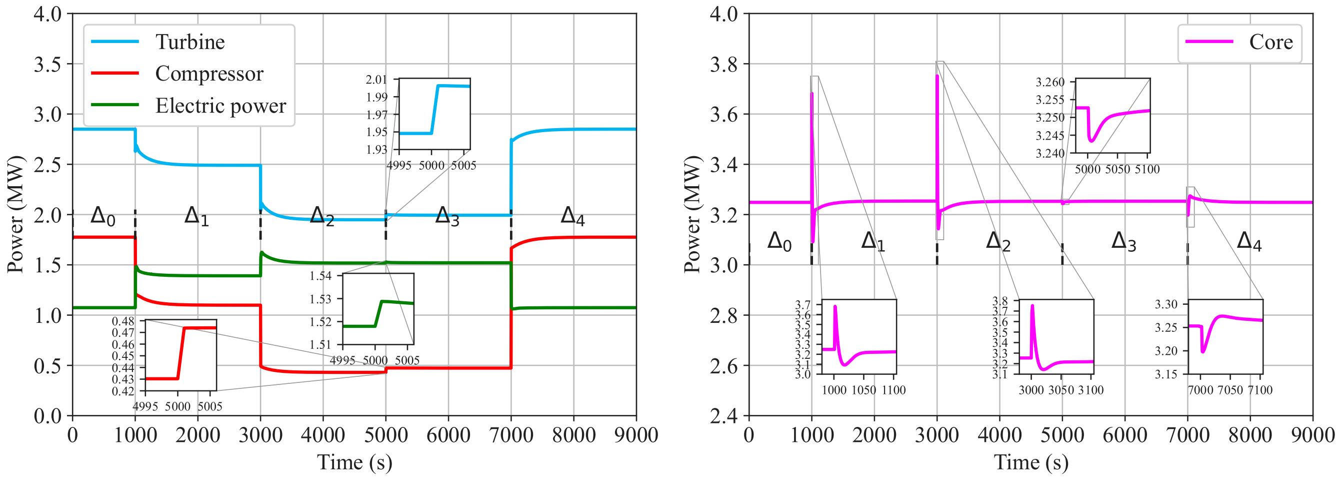
Condition 3 (*Fluctuating Working Fluid Flow*)**.** As shown in Fig. 35, the impact of changes in the mass flow rate of He-Xe gas in the CBC on the system power is considered. The mass flow rate of He-Xe gas is set to change from 13 kg/s during the Δ_0 period to 12.5 kg/s during the Δ_1 period. After running for 2000 s, it then changes to 12 kg/s during the Δ_2 period. After another 2000 s of operation, the mass flow rate returns to 12.5 kg/s during the Δ_3 period. Finally, it returned to 13 kg/s during Δ_4.

As illustrated in Fig. 36(a), when the He-Xe gas mass flow rate decreases from 13 kg/s to 12.5 kg/s or 12 kg/s, the output power of the turbine exhibits a decreasing trend, while the compressor power increases, leading to a reduction in net power generation. Conversely, as the mass flow rate rises from 12 kg/s to 12.5 kg/s or recovers to 13 kg/s, the turbine power increases accordingly, accompanied by a reduction in compressor power, thereby enhancing the overall power output. This behavior can be fundamentally explained with reference to the characteristic curves of the turbine and compressor shown in Figs. 8 and 9, as well as Eqs. (17) and (20). Under constant rotational speed, variations in mass flow rate have a more significant impact on the expansion ratio of the turbine than on the compression ratio of the compressor, which remains relatively stable. Consequently, when the mass flow rate decreases, the flow velocity within the compressor is reduced, leading to weakened cooling performance and an increased temperature rise during the compression process. This results in higher specific compression work, thereby increasing compressor power consumption and reducing overall efficiency. Simultaneously, the reduced expansion ratio in the turbine limits the available enthalpy drop for expansion. Although the temperature rise remains relatively unchanged, the reduced mass flow rate leads to a decline in turbine output power. The absolute value of the increased temperature difference between the turbine and compressor inlet and outlet ($|T_{out} - T_{in}| - |T'_{out} - T'_{in}|$, T' represents the inlet and outlet temperatures of the turbine or compressor under steady-state conditions.) during this process is depicted in Fig. 37(a).

As the mass flow rate of the He-Xe gas decreases, both the average core temperature and outlet temperature of the reactor increase significantly due to the reduction in cooling capacity, as illustrated in Fig. 37(b). Initially, this thermal accumulation causes a transient rise in reactor power, as the generated heat momentarily exceeds the system's heat removal capability. However, the elevated fuel temperature activates strong negative reactivity feedback mechanisms, as shown in Eq. (4). After reaching a short-lived peak, the core power output begins to decrease and eventually stabilizes at a level below the original steady-state value. This behavior highlights the reactor's inherent self-regulating nature under thermal disturbances. Moreover, the low thermal inertia and heat capacity of the He-Xe mixture exacerbate the temperature rise while limiting effective heat removal, further contributing to the decline in core power. Conversely, when the He-Xe mass flow rate increases, both the core and outlet temperatures drop significantly. The reactor power in this case exhibits an initial decrease followed by a rebound above the original value, also governed by reactivity feedback mechanisms, as shown in Fig. 36(b). These observed dynamics demonstrate a realistic coupling between thermal-hydraulic and neutronic effects, validating the model's accuracy under off-design operational conditions.

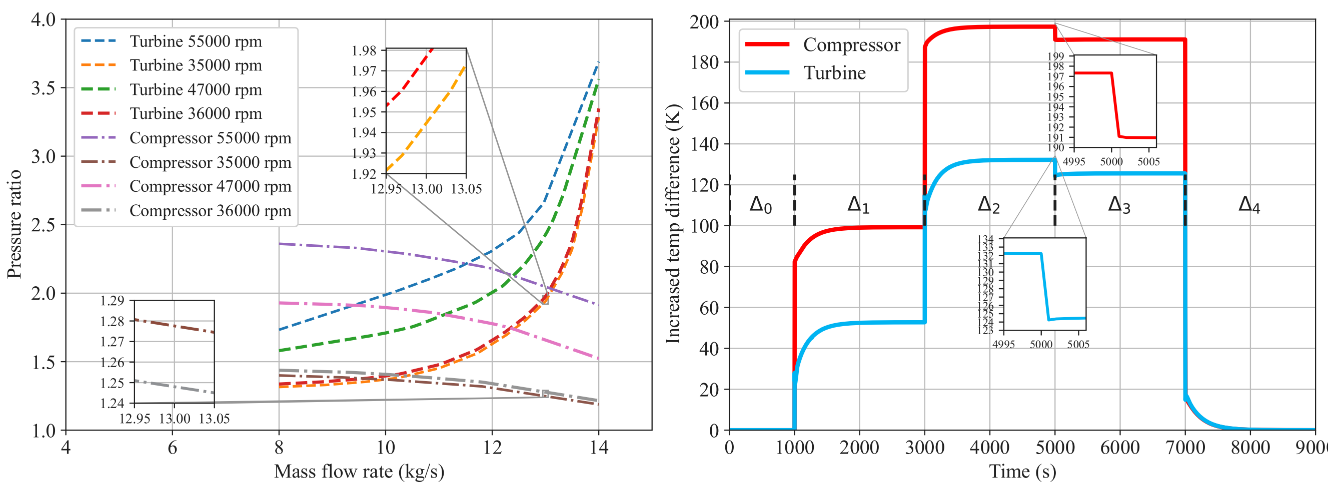
Condition 4 (*Changing Load*)**.** In this operating condition, the load is first maintained at the original steady state of 1072 kW for 100 s, then reduced to 1050 kW for 500 s. After that, it is increased to 1200 kW for another 500 s, before being reduced to 800 kW for a further 500 s, as illustrated in Fig. 38(a).

The most immediate impact of load variation is the change in the shaft speed of the TAC. As shown in Fig. 38(b), when the electrical



(a) TAC power (b) Core power

Fig. 33. Power variations under changes in shaft speed.



(a) Turbine & Compressor Curves (b) Temp Difference

Fig. 34. Immediate cause of Power variations under changes in shaft speed.

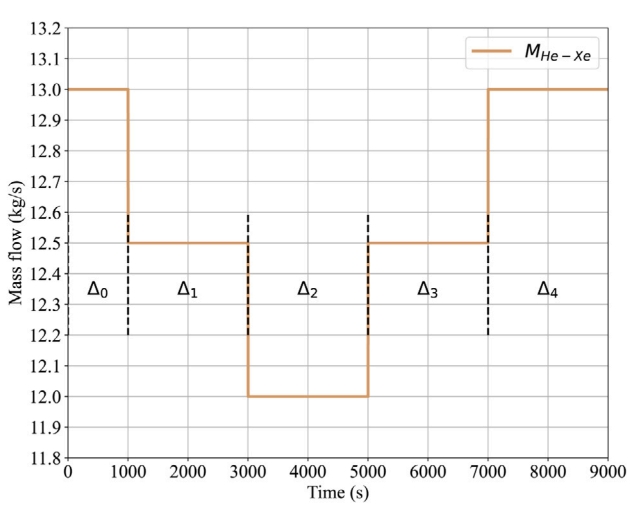
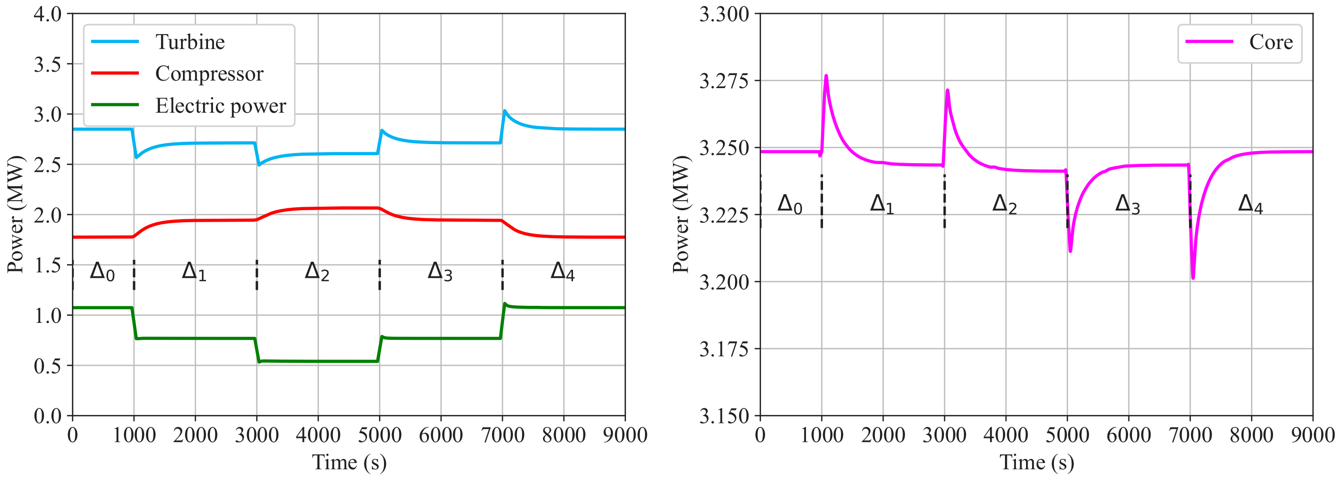


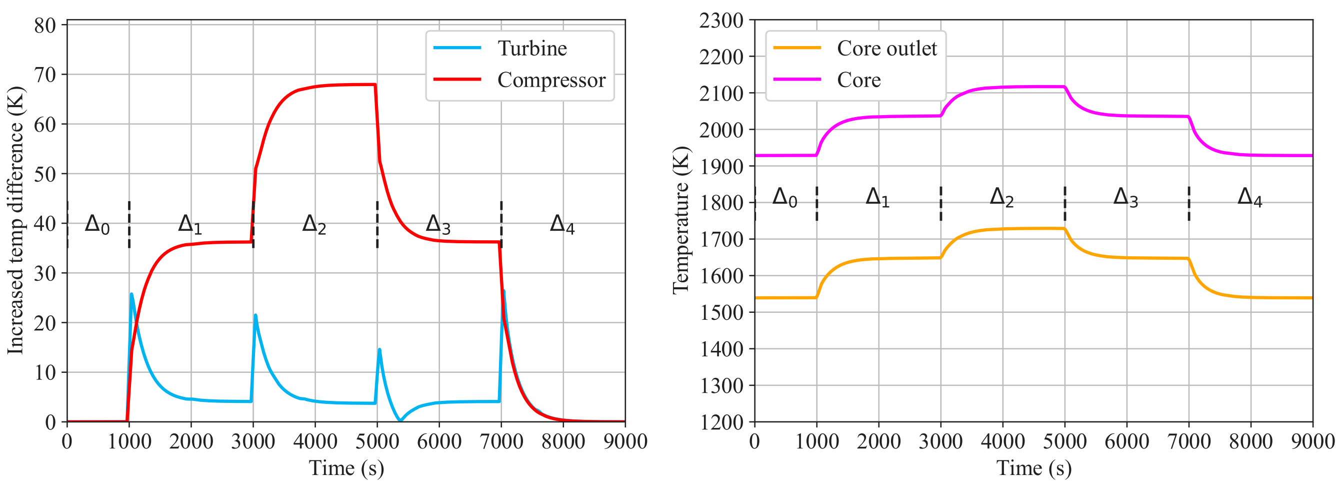
Fig. 35. Variations in working fluid flow.

load decreases to 1050 kW, the shaft speed increases to 58987 rpm, owing to the initially positive excess power, as indicated at the end of phase Δ_0 in Fig. 38(e). With the rising shaft speed, both the compressor and turbine power increase. However, since the enthalpy rise of the compressor exceeds that of the turbine, the net electric power output decreases to 1050 kW, as shown in phase Δ_1 of Fig. 38(c). At this stage, the excess load becomes zero, and the shaft speed stabilizes at 58314 rpm. Moreover, during this phase, the core temperature and core power initially decrease. However, due to the reactivity feedback mechanism, together with the increased enthalpy changes in the turbine and compressor, the core temperature and outlet temperature gradually rise. Eventually, the reactor power stabilizes at 3.249 MW, as depicted in Fig. 38(d), phase Δ_1. In contrast, when the load increases to 1200 kW, the shaft speed correspondingly drops, accompanied by a decrease in the power of both the turbine and compressor. Since the coolant flow remains constant, the electric power increases, and the reactor power temporarily rises before decreasing to 3.244 MW. As a result, the overall thermal efficiency improves. A similar trend is observed when the load increases further to 1500 kW in phase Δ_3. It is worth noting that the response time for shaft speed adjustment during the 1500 kW load phase is significantly longer than that in Δ_2. This is consistent with the results from Condition 2, where the enthalpy rise of the compressor and turbine do not always exhibit a fixed relationship,



(a) TAC power (b) Core power

Fig. 36. Power variations under changes in working fluid flow.



(a) Turbine & compressor (b) Core & outlet

Fig. 37. Temperature variations under changes in working fluid flow.

resulting in nonlinear changes in net electric power. To maintain power balance, the shaft speed undergoes self-regulating oscillations. When the load reaches 1500 kW, which far exceeds the design value of 1072 kW, a longer adjustment time is required. Finally, when the load drops to 800 kW, the shaft speed reaches a maximum of 60000 rpm. As the He-Xe coolant flow remains constant and the compressor and turbine performance reach their limits, the TAC's excess power remains at approximately 200 kW, as shown in phase Δ_4 of Fig. 38(e). This causes a continuous accumulation of heat that cannot be effectively rejected, leading to a sustained rise in core temperature, which poses significant safety concerns.

In summary, the experimental results are consistent with theoretical expectations, and they offer valuable insights for enhancing core performance and investigating operational safety thresholds.

4. Conclusion

This paper investigates the Space Nuclear Reactor Closed Brayton Cycle (SNRCBC) system, proposing a universal research framework. The input parameters in the development process of the SNRCBC system model are categorized into static and dynamic parameters. Corresponding models are established based on physical principles, and experimental results are obtained by incorporating design parameters. These results are then analyzed to elucidate the intrinsic relationship between system characteristics and input parameters. The research involves the development of core models, a co-axial turbine-generator-compressor integration model, heat exchange models (including heaters, recuperators, coolers, and radiators), and material property packages that represent the physical characteristics of materials and working fluids. Each component is studied individually and in combination, with a startup scheme proposed and a dynamic simulation program written. Additionally, the dynamic response of the system under various operating conditions is examined. The research findings and results are summarized as follows:

1. Under the designed steady-state operating conditions, the output power of the core is 3.248 MW, the output power of the turbine is 2.849 MW, the output power of the compressor is 1.777 MW, the power generated is 1.072 MW, and the thermoelectric conversion efficiency is 33.03%. By calibrating the steady-state model, the transient startup process was designed in eight stages, each logically coherent. The computed results after reaching steady state were found to align well with the design values, thus validating the correctness of the system model and startup scheme established in this paper.

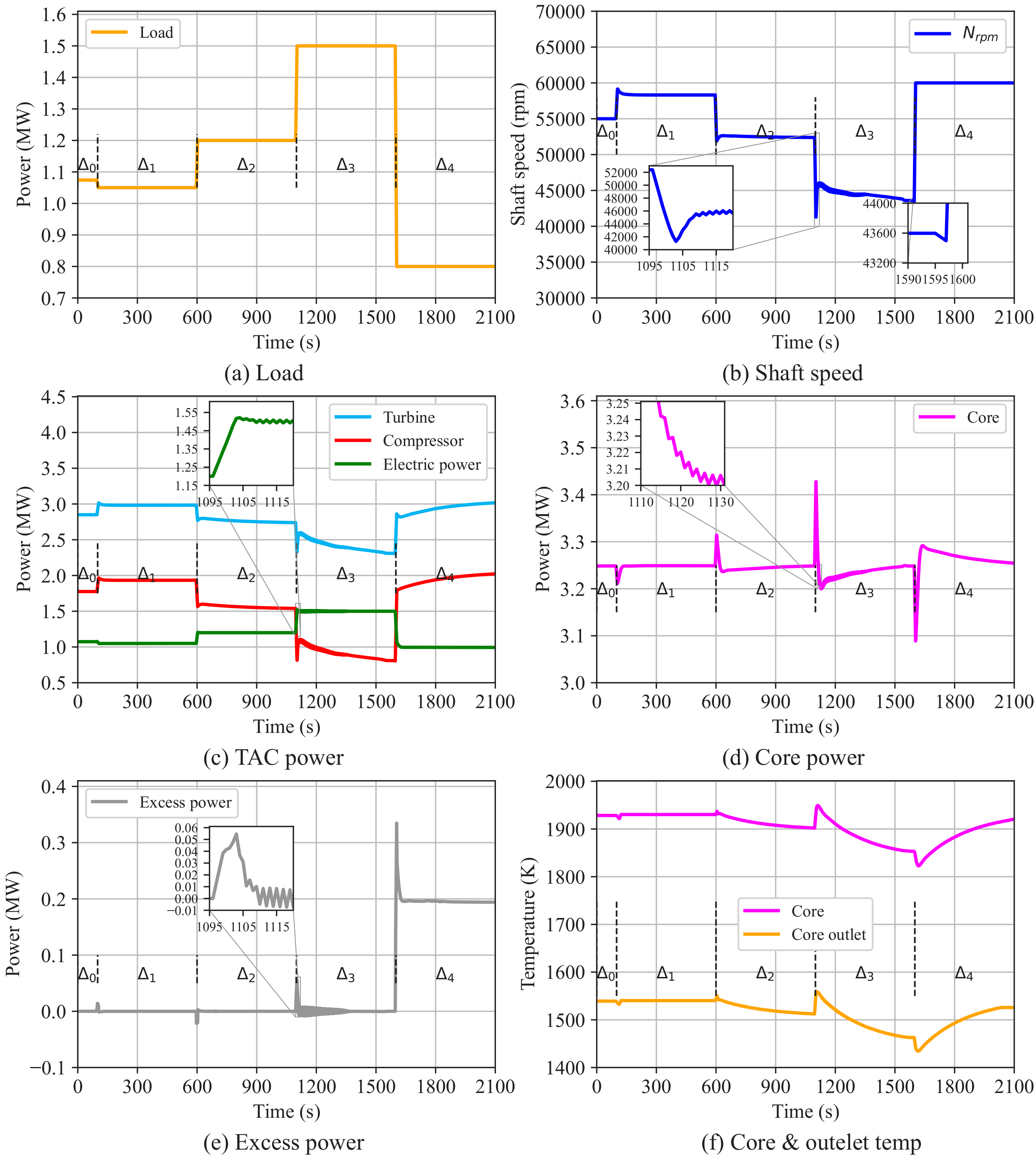


Fig. 38. System response changes in changing load.

2. Due to the co-axial design of the turbine, generator, and compressor, the system exhibits strong self-regulation under varying reactivity, shaft speed, fluid flow, and load conditions. The model remains stable under transient conditions considered in this study. An increase in reactivity raises core power and temperature, boosting turbine and compressor output, thus increasing power generation. A decrease in shaft speed reduces turbine and compressor power, The reduction in rotor speed will decrease the power of the turbine and compressor. However, when the mass flow rate of the He-Xe coolant remains constant, the power generation capacity shows a non-linear response, which mainly depends on the enthalpy changes of the turbine and compressor. A decrease in working fluid flow yields similar results. A reduction in load accelerates the shaft, while an increase in load results in a decrease in shaft speed.

Although this work provides a detailed introduction to the development of the SNRCBC model, offering relevant parameters, analyzing the system's steady-state operation, proposing a startup plan, validating the model's accuracy, and conducting simulation analysis under four different operating conditions — varying reactivity, shaft speed, working fluid flow rate, and load — the research on SNRCBC remains relatively rudimentary. Future work could build upon this foundation to optimize the overall system structure and parameter selection, and conduct further performance analysis, thereby continuously improving power generation efficiency and reducing costs.

CRediT authorship contribution statement

Xiangrong Tang: Writing – review & editing, Writing – original draft, Project administration, Methodology, Conceptualization. **Zongxin Yang:** Software, Formal analysis, Data curation, Conceptualization. **ChuangKan Liu:** Software, Formal analysis, Data curation, Conceptualization. **Linyu Huang:** Writing – review & editing, Supervision. **Haosen Liu:** Formal analysis, Data curation. **Qian Ning:** Validation, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Virial coefficient of He-Xe mixture

The second-order Virial coefficient $\overline{B}$ for a mixture of gases is calculated as follows:

$$\overline{B} = x_1^2 \cdot B_{11} + 2x_1 \cdot x_2 \cdot B_{12} + x_2^2 \cdot B_{22} \tag{A.1.1}$$

The subscripts 1 and 2 represent helium and xenon, respectively. B_{11} is the second-order Virial coefficient of helium, B_{22} is the second-order Virial coefficient of xenon, B_{12} is the second-order cross Virial coefficient for the He-Xe gas mixture, x_1 is the mole fraction of helium, and x_2 is the mole fraction of xenon.

$$B_{11} = (8.4 - 0018T + \frac{115}{\sqrt{T}} - \frac{835}{T}) \times 10^{-6} \tag{A.1.2}$$

$$B_{22} = V_2^* \cdot [-102.6 + (102.732 - 0.01\theta_2 - \frac{0.44}{\theta^1.22_2}) \cdot \tanh(4.5\sqrt{\theta_2})] \tag{A.1.3}$$

$$B_{12} = V_{12}^* \cdot [-102.6 + (102.732 - 0.01\theta_{12} - \frac{0.44}{\theta^1.22_{12}}) \cdot \tanh(4.5\sqrt{\theta_{12}})] \tag{A.1.4}$$

where, V_2^* and V_{12}^* represent the characteristic molar volumes of xenon and the He-Xe gas mixture, respectively, while θ_2 and θ_{12} denote the reduced temperatures of xenon and the He-Xe gas mixture, respectively. For the third-order Virial coefficient $\overline{C}$ of the gas mixture, its calculation follows the same approach as that of the third-order Virial coefficients of helium and xenon:

$$\overline{C} = x_1^3 C_{111} + 3x_1^2 x_2 C_{112} + 3x_1 x_2^2 C_{122} + x_2^3 C_{222} \tag{A.1.5}$$

where C_{111} is the third-order Virial coefficient of helium, C_{222} is the third-order Virial coefficient of xenon, and C_{112} and C_{122} are the third-order cross-Virial coefficients of the He-Xe gas mixture. The relations are given by $C_{112} = (C_{111}^2 C_{222})^{1/3}$ and $C_{122} = (C_{111} C_{222}^2)^{1/3}$. The third-order Virial coefficients of pure gases can be obtained using the formula (A.1.6):

$$C_{iii} = V_i^* \cdot [-0.0862 - 3.6 \times 10_{-5}\theta_i + \frac{0.0237}{\theta_i^{0.059}} \cdot \tanh(0.84\theta_i)] \tag{A.1.6}$$

A.2. The dynamic viscosity of the He-Xe gas mixture

The viscosity of binary gas mixtures as:

$$\overline{\mu}(T,P) = \overline{\mu^0}(T) + (1 - \frac{1}{2.3}) \cdot x_2 \cdot \overline{\mu}^* \cdot \Psi_\mu(\rho_r) \tag{A.2.1}$$

μ_0 represents the viscosity of the He-Xe gas mixture under ideal gas conditions, calculated using the Chapman–Enskog kinetic approach as:

$$\overline{\mu}^0 = \mu_1^0/(1 + \phi_{12}x_2/x_1) + \mu_2^0/(1 + \phi_{21}x_1/x_2) \tag{A.2.2}$$

$$\phi_{12} = \frac{\mu_1^0}{\mu_{12}}[\frac{2m_1m_2}{(m_1+m_2)^2}] \times [\frac{5}{3A_{12}^*} + \frac{m_2}{m_1}] \tag{A.2.3}$$

$$\phi_{21} = \frac{\mu_2^0}{\mu_{12}}[\frac{2m_1m_2}{(m_1+m_2)^2}] \times [\frac{5}{3A_{12}^*} + \frac{m_1}{m_2}] \tag{A.2.4}$$

$$\mu_{12} = A_\mu (T - T_\mu)^n \tag{A.2.5}$$

where, m_1 and m_2 represent the molecular masses of helium and xenon, respectively. u_1^0 and u_2^0 denote the viscosities of helium and xenon, respectively, under ideal conditions. The coefficients $A_\mu = 3.40998 \times 10^{-7}$, $T_\mu = 45.89$ K, $n = 0.658754$, $1.094 < A_{12}^* < 1.119$.

The "pseudo-critical" viscosity of the mixture, $\overline{\mu}^*$, is calculated as:

$$\overline{\mu}^* = 0.204 \times 10^{-7} \frac{\sqrt{\overline{M} \cdot \overline{T}^*}}{(0.291 \times \overline{V}^*)^{2/3}} \tag{A.2.6}$$

$$\overline{M} = x_1 M_1 + x_2 M_2 \tag{A.2.7}$$

$$\overline{V}^* = x_1 V_{11}^* + x_2 V_{22}^* \tag{A.2.8}$$

$$V_{ii}^* = R_g (T_{cr}/R_{cr}) \tag{A.2.9}$$

R_g is the gas constant, while T_{cr} and P_{cr} denote the critical temperature and critical pressure, respectively. The $\Psi_\mu(\rho_r)$ is a viscosity function with density as the variable:

$$\Psi_\mu(\rho_r) = 0.221\rho_r + 1.062\rho_r^2 - 0.509\rho_r^3 + 0.225\rho_r^4 \tag{A.2.10}$$

$$\rho_r = \rho/\rho_{cr} \tag{A.2.11}$$

Here, ρ_r is reduced density, ρ represents the gas density, and ρ_{cr} refers to the critical density.

A.3. The thermal conductivity of the He-Xe gas mixture

For the He-Xe gas mixture, its thermal conductivity is given by the semi-empirical formula:

$$\overline{\lambda} = \overline{\lambda}^0 + (1 - \frac{1}{2.94})\overline{\lambda}_{cr}^* \cdot \Psi_\lambda(\rho_r) \tag{A.3.1}$$

$$\Psi_\lambda(\rho_r) = 0.645\rho_r + 0.331\rho_r^2 - 0.0368\rho_r^3 - 0.0128\rho_r^4 \tag{A.3.2}$$

where, $\Psi_\lambda(\rho_r)$ is the thermal conductivity function with density as a variable, and $\overline{\lambda}^0$ represents the thermal conductivity in an ideal state:

$$\overline{\lambda}^0 = (\frac{x_1^2}{L_{11}} - \frac{2x_1x_2 \cdot L_{12}}{L_{11}L_{22}}) \cdot (1 - \frac{L_{12}^2}{L_{11}L_{22}})^{-1} \tag{A.3.3}$$

Table A.5
Fundamental and critical properties of He and Xe.

Property	Helium	Xenon
M (kg/mole)	0.004003	0.13129
T_{cr} (K)	5.2	289.7
P_{cr} (MPa)	0.2275	5.84
ρ_{cr} (kg/m^2)	69.64	1110
V_{cr} (cm^3/mole)	57.48	118.28
$0.291V^*$ (cm^3/mole)	55.30	120.02
V^* (cm^3/mole)	190.04	412.45
$A_\mu \times 10^7$	3.0629	7.5683
T_μ (K)	−21.33	112.31
$\mu_0(T_{cr}) \times 10^6$ (Pa s)	1.428	22.55
$\lambda^0(T_{cr}) \times 10^3$ (W/m K)	10.32	5.416
$\lambda_{cr} \times 10^3$ (W/m K)	34.32	15.966

$$L_{ii} = \frac{x_i^2}{\lambda_i^0} + \frac{x_i x_j}{2\lambda_{ij}} \frac{7.5m_i^2 + 6.25m_j^2 - 3m_j^2 \cdot B_{ij}^* + 4m_i m_j \cdot A_{ij}^*}{(m_i + m_j)^2 \cdot A_{ij}^*} \tag{A.3.4}$$

$$L_{ij} = -\frac{x_i x_j}{2\lambda_{ij}} \frac{m_1 m_2}{(m_1 + m_2)^2 \cdot A_{12}^*} (\frac{55}{4} - 3B_{12}^* - 4A_{12}^*) \tag{A.3.5}$$

The relevant parameters for helium and xenon are presented in Table A.5.

A.4. The physical properties of the metals

The formula for calculating the physical properties of liquid lithium is as follows:

$$\lambda[\mathrm{W/(m\ K)}] = 24.8 + 45 \times 10^{-3} \cdot T - 11.6 \times 10^{-6} \cdot T^2 \tag{A.4.1}$$

$$\rho\ (\mathrm{kg/m^3}) = 0.53799943 - 0.016043986 \cdot (T \times 10^{-3}) - 0.099963362 \cdot (T \times 10^{-3})^2 + 0.054609894 \cdot (T \times 10^{-3})^3 - 0.015087628 \cdot (T \times 10^{-3})^4 + 0.0027045593 \cdot (T \times 10^{-3})^5 - 0.00031537739 \cdot (T \times 10^{-3})^6 \tag{A.4.2}$$

$$C_p\ [\mathrm{J/(kg\ K)}] = (31.227 + 0.205 \times 10^6 T^{-2} - 5.265 \times 10^{-3} T + 2.628 \times 10^{-6} T^2)/6.941 \tag{A.4.3}$$

$$\ln \mu\ (\mathrm{Pa\ s}) = -4.16435 - 0.63740 \ln T + 292.1/T \tag{A.4.4}$$

where λ is the thermal conductivity, ρ is the density, C_p is the specific heat capacity at constant pressure, μ is the dynamic viscosity, and T is the temperature.

The formula for calculating the physical properties of 22Na-78K is as follows:

$$1/\rho_{\mathrm{NaK}} = 0.22/\rho_{\mathrm{Na}} + 0.78/\rho_{\mathrm{K}} \tag{A.4.5}$$

$$C_p\ [\mathrm{J/(kg\ K)}] = 0.22C_p(\mathrm{Na}) + 0.78C_p(\mathrm{K}) \tag{A.4.6}$$

$$\lambda\ [\mathrm{W/(m\ K)}] = 15.0006 + 30.2877 \times 10^{-3} T - 20.8095 \times 10^{-6} T^2 \tag{A.4.7}$$

$$\nu \cdot 10^8 (\mathrm{m^2/s}) = 200.7657 - 0.734683 \cdot T + 1.12102 \times 10^{-3} \cdot T^2 - 0.774427 \times 10^{-6} \cdot T^3 + 0.200382 \times 10^{-9} \cdot T^4 \tag{A.4.8}$$

Here, ρ_{Na} and ρ_K represent the densities of sodium (Na) and potassium (K) respectively, C_p denotes the specific heat capacity at constant pressure, λ is the thermal conductivity, and ν refers to the dynamic viscosity.

The formula for calculating the physical properties of austenite stainless steel type 316 is as follows:

$$\rho\ (\mathrm{kg/m^3}) = 8084 - 0.4209T - 3.894 \times 10^{-5} T^2 \tag{A.4.9}$$

$$C_p\ [\mathrm{J/(kg\ K)}] = 462 + 0.134T \tag{A.4.10}$$

$$\lambda\ [\mathrm{W/(m\ K)}] = 9.248 + 0.01571T \tag{A.4.11}$$

where ρ is the density, C_p is the specific heat capacity at constant pressure, and λ is the thermal conductivity.

The formula for calculating the physical properties of Aluminium type 316 is as follows:

$$\rho = 2770 \cdot (1 - 0.00021 \cdot (T - 300)) \tag{A.4.12}$$

$$\lambda = 235 - 0.07 \cdot (T - 300) \tag{A.4.13}$$

$$C_p = 900 + 0.5 \cdot (T - 300) \tag{A.4.14}$$

where ρ is the density, λ is the thermal conductivity, and C_p is the specific heat capacity at constant pressure.

The formula for calculating the physical properties of Nickel-based alloy is as follows:

$$\lambda = 0.015536T + 14.367857 \tag{A.4.15}$$

$$\rho = -0.4T + 8309.2 \tag{A.4.16}$$

$$C_p = 418.8 + 0.2615T - 4.598 \times 10^{-4} T^2 + 2.365 \times 10^{-7} T^3 \tag{A.4.17}$$

where λ is the thermal conductivity, ρ is the density, and C_p is the specific heat capacity at constant pressure.

Data availability

No data was used for the research described in the article.

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